SUSE Linux Enterprise Server 15 SP5 (prerelease)

Release Notes

SUSE Linux Enterprise Server is a modern, modular operating system for both multimodal and traditional IT. This document provides a high-level overview of features, capabilities, and limitations of SUSE Linux Enterprise Server 15 SP5 (prerelease) and highlights important product updates.

This product will be released in June 2023. The latest version of these release notes is always available at https://www.suse.com/releasenotes. Drafts of the general documentation can be found at https://susedoc.github.io/doc-sle/main.

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1 About the release notes

These Release Notes are identical across all architectures, and the most recent version is always available online at https://www.suse.com/releasenotes.

Entries are only listed once but they can be referenced in several places if they are important and belong to more than one section.

Release notes usually only list changes that happened between two subsequent releases. Certain important entries from the release notes of previous product versions are repeated. To make these entries easier to identify, they contain a note to that effect.

However, repeated entries are provided as a courtesy only. Therefore, if you are skipping one or more service packs, check the release notes of the skipped service packs as well. If you are only reading the release notes of the current release, you could miss important changes.

2 SUSE Linux Enterprise Server

SUSE Linux Enterprise Server 15 SP5 (prerelease) is a multimodal operating system that paves the way for IT transformation in the software-defined era. It is a modern and modular OS that helps simplify multimodal IT, makes traditional IT infrastructure efficient and provides an engaging platform for developers. As a result, you can easily deploy and transition business-critical workloads across on-premises and public cloud environments.

SUSE Linux Enterprise Server 15 SP5 (prerelease), with its multimodal design, helps organizations transform their IT landscape by bridging traditional and software-defined infrastructure.

2.1 Interoperability and hardware support

Designed for interoperability, SUSE Linux Enterprise Server integrates into classical Unix and Windows environments, supports open standard interfaces for systems management, and has been certified for IPv6 compatibility.

This modular, general-purpose operating system runs on four processor architectures and is available with optional extensions that provide advanced capabilities for tasks such as real-time computing and high-availability clustering.

SUSE Linux Enterprise Server is optimized to run as a high-performance guest on leading hypervisors. This makes SUSE Linux Enterprise Server the perfect guest operating system for virtual computing.

2.2 What is new?

2.2.1 General changes in SLE 15

SUSE Linux Enterprise Server 15 introduces many innovative changes compared to SUSE Linux Enterprise Server 12. The most important changes are listed below.

Migration from openSUSE Leap to SUSE Linux Enterprise Server

SLE 15 SP2 and later support migrating from openSUSE Leap 15 to SUSE Linux Enterprise Server 15. Even if you decide to start out with the free community distribution, you can later easily upgrade to a distribution with enterprise-class support. For more information, see the *Upgrade Guide* at https://documentation.suse.com/sles/15-SP5/html/SLES-all/cha-upgrade-online.html#sec-upgrade-online-opensuse-to-sle .

Extended package search

Use the new Zypper command `zypper search-packages` to search across all SUSE repositories available for your product, even if they are not yet enabled. For more information see *Section 5.12.6, "Searching packages across all SLE modules"*.

Software Development Kit

In SLE 15, packages formerly shipped as part of the Software Development Kit are now integrated into the products. Development packages are packaged alongside other packages. In addition, the *Development Tools* module contains tools for development.

RMT replaces SMT

SMT (Subscription Management Tool) has been removed. Instead, RMT (Repository Mirroring Tool) now allows mirroring SUSE repositories and custom repositories. You can then register systems directly with RMT. In environments with tightened security, RMT can also proxy other RMT servers. If you are planning to migrate SLE 12 clients to version 15, RMT is the supported product to handle such migrations. If you still need to use SMT for these migrations, beware that the migrated clients will have *all* installation modules enabled. For more information see *Section 4.2.5, "SMT has been replaced by RMT"*.

Media changes

The *Unified Installer* and *Packages* media known from SUSE Linux Enterprise Server 15 SP1 have been replaced by the following media:

- **Online Installation Medium:** Allows installing all SUSE Linux Enterprise 15 products. Packages are fetched from online repositories. This type of installation requires a registration key. Available SLE modules are listed in *Section 3.1, "Modules in the SLE 15 SP5 (prerelease) product line"*.

- **Full Installation Medium:** Allows installing all SUSE Linux Enterprise Server 15 products without a network connection. This medium contains all packages from all SLE modules. SLE modules need to be enabled manually during installation. RMT (Repository Mirroring Tool) and SUSE Manager provide additional options for disconnected or managed installations.

MAJOR UPDATES TO THE SOFTWARE SELECTION:

Salt

SLE 15 SP5 (prerelease) can be managed via Salt, making it integrate better with modern management solutions such as SUSE Manager.

Python 3

As the first enterprise distribution, SLE 15 offers full support for Python 3 development in addition to Python 2.

Directory Server

389 Directory Server replaces OpenLDAP as the LDAP directory service. OpenLDAP has been re-added to SLE 15 SP5 and SLE SP6 to prolong the time window for a migration to 389 Directory Server. Support for OpenLDAP on SLE 15 will end with the SLE 15 SP6 lifecycle. Future versions and service packs of SLE will not include OpenLDAP.

2.2.2 Changes in 15 SP5 (prerelease)

SUSE Linux Enterprise Server 15 SP5 (prerelease) introduces changes compared to SUSE Linux Enterprise Server 15 SP4. The most important changes are listed below:

2.2.3 Package and module changes in 15 SP5 (prerelease)

The full list of changed packages compared to 15 SP4 can be seen at this URL:

- https://documentation.suse.com/package-lists/sle/15-SP5/package-changes_SLE-15-SP4-GA_SLE-15-SP5-GA.txt

The full list of changed modules compared to 15 SP4 can be seen at this URL:

- https://documentation.suse.com/package-lists/sle/15-SP5/module-changes_SLE-15-SP4-GA_SLE-15-SP5-GA.txt

2.3 Important sections of this document

If you are upgrading from a previous SUSE Linux Enterprise Server release, you should review at least the following sections:

- *Section 2.7, "Support statement for SUSE Linux Enterprise Server"*
- *Section 4.2, "Upgrade-related notes"*
- *Section 5, "Changes affecting all architectures"*

2.4 Security, standards, and certification

SUSE Linux Enterprise Server 15 SP5 (prerelease) will not be submitted for certification to either Common Criteria or NIST FIPS 140-3. As a practice, SUSE only submits even-numbered Service Packs (for example: SLES 15, SP2, SP4, etc.) for certification.

For more information about certification, see https://www.suse.com/support/security/certifications/ .

2.5 Documentation and other information

2.5.1 Available on the product media

- Read the READMEs on the media.
- Get the detailed change log information about a particular package from the RPM (where `FILENAME.rpm` is the name of the RPM):

```
rpm --changelog -qp FILENAME.rpm
```

- Check the `ChangeLog` file in the top level of the installation medium for a chronological log of all changes made to the updated packages.
- Find more information in the `docu` directory of the installation medium of SUSE Linux Enterprise Server 15 SP5 (prerelease). This directory includes PDF versions of the SUSE Linux Enterprise Server 15 SP5 (prerelease) Installation Quick Start Guide.

2.5.2 Online documentation

- For the most up-to-date version of the documentation for SUSE Linux Enterprise Server 15 SP5 (prerelease), see https://susedoc.github.io/doc-sle/main (draft version).
- Find change logs for public cloud images at the Public Cloud Information Tracker (PINT) at https://pint.suse.com/.

2.6 Support and life cycle

SUSE Linux Enterprise Server is backed by award-winning support from SUSE, an established technology leader with a proven history of delivering enterprise-quality support services.

SUSE Linux Enterprise Server 15 has a 13-year life cycle, with 10 years of General Support and three years of Extended Support. The current version (SP5) will be fully maintained and supported until six months after the release of SUSE Linux Enterprise Server 15 SP6.

If you need additional time to design, validate and test your upgrade plans, Long Term Service Pack Support can extend the support duration. You can buy an additional 12 to 36 months in twelve month increments. This means that you receive a total of three to five years of support per Service Pack.

For more information, see the pages Support Policy (https://www.suse.com/support/policy.html) and Long Term Service Pack Support (https://www.suse.com/support/programs/long-term-service-pack-support.html).

2.7 Support statement for SUSE Linux Enterprise Server

To receive support, you need an appropriate subscription with SUSE. For more information, see https://www.suse.com/support/?id=SUSE_Linux_Enterprise_Server.

The following definitions apply:

L1

Problem determination, which means technical support designed to provide compatibility information, usage support, ongoing maintenance, information gathering, and basic troubleshooting using the documentation.

L2

Problem isolation, which means technical support designed to analyze data, reproduce customer problems, isolate the problem area, and provide a resolution for problems not resolved by Level 1 or prepare for Level 3.

L3

Problem resolution, which means technical support designed to resolve problems by engaging engineering to resolve product defects which have been identified by Level 2 Support.

For contracted customers and partners, SUSE Linux Enterprise Server is delivered with L3 support for all packages, except for the following:

- Technology Previews, see *Section 2.8, “Technology previews”*
- Sound, graphics, fonts and artwork
- Packages that require an additional customer contract, see *Section 2.7.2, “Software requiring specific contracts”*
- Some packages shipped as part of the module *Workstation Extension* are L2-supported only
- Packages with names ending in `-devel` (containing header files and similar developer resources) will only be supported together with their main packages.

SUSE will only support the usage of original packages. That is, packages that are unchanged and not recompiled.

2.7.1 General support

To learn about supported features and limitations, refer to the following sections in this document:

- *Section 5.6, "Kernel"*
- *Section 5.10, "Storage and file systems"*
- *Section 5.13, "Virtualization"*
- *Section 9, "Removed and deprecated features and packages"*

2.7.2 Software requiring specific contracts

Certain software delivered as part of SUSE Linux Enterprise Server may require an external contract. Check the support status of individual packages using the RPM metadata that can be viewed with `rpm`, `zypper`, or YaST.

Major packages and groups of packages affected by this are:

- PostgreSQL (all versions, including all subpackages)

2.7.3 Software under GNU AGPL

SUSE Linux Enterprise Server 15 SP5 (prerelease) (and the SUSE Linux Enterprise modules) includes the following software that is shipped *only* under a GNU AGPL software license:

- Ghostscript (including subpackages)

SUSE Linux Enterprise Server 15 SP5 (prerelease) (and the SUSE Linux Enterprise modules) includes the following software that is shipped under multiple licenses that include a GNU AGPL software license:

- MySpell dictionaries and LightProof
- ArgyllCMS

2.8 Technology previews

Technology previews are packages, stacks, or features delivered by SUSE to provide glimpses into upcoming innovations. Technology previews are included for your convenience to give you a chance to test new technologies within your environment. We would appreciate your feedback! If you test a technology preview, contact your SUSE representative and let them know about your experience and use cases. Your input is helpful for future development.

Technology previews come with the following limitations:

- Technology previews are still in development. Therefore, they may be functionally incomplete, unstable, or in other ways not suitable for production use.
- Technology previews are **not** supported.
- Technology previews may only be available for specific hardware architectures. Details and functionality of technology previews are subject to change. As a result, upgrading to subsequent releases of a technology preview may be impossible and require a fresh installation.
- Technology previews can be removed from a product at any time. This may be the case, for example, if SUSE discovers that a preview does not meet the customer or market needs, or does not comply with enterprise standards.

2.8.1 Technology previews for Arm 64-Bit (AArch64)

2.8.1.1 KVM virtualization with 64K page size kernel flavor

As a technology preview, SUSE Linux Enterprise Server for Arm 15 SP3 added a kernel flavor `64kb`. SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) introduces support for this `64kb` kernel flavor (*Section 8.3, “64K page size kernel flavor is supported”*).

KVM virtualization with this `64kb` kernel flavor remains a technology preview. Use the `default` kernel flavor for virtualization support.

2.8.1.2 Driver enablement for NVIDIA BlueField-2 DPU as host platform

SUSE Linux Enterprise Server for Arm 15 SP1 and later kernels include drivers for installing on NVIDIA* BlueField* Data Processing Unit (DPU) based server platforms and SmartNIC (Network Interface Controller) cards.

As a technology preview, the SUSE Linux Enterprise Server for Arm 15 SP4 and SP5 kernels include drivers for running on NVIDIA BlueField-2 DPU.

Should you wish to use SUSE Linux Enterprise Server for Arm on NVIDIA BlueField-2 or BlueField-2X (or BlueField-3) in production, contact your SUSE representative.

Note: Host drivers and tools for NVIDIA BlueField-2 SmartNICs

This Technology Preview status applies only to installing SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) **on** NVIDIA BlueField-2 DPUs.

For an NVIDIA BlueField-2 DPU PCIe card inserted as **SmartNIC** into a SUSE Linux Enterprise Server 15 SP5 (prerelease) or SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) based server, check *Section 2.8, "Technology previews"* and *Section 5.6, "Kernel"* for support status or known limitations of NVIDIA ConnectX* network drivers for BlueField-2 DPUs (`mlx5_core` and others).

The `rshim` tool is available from SUSE Package Hub (*Section 5.11, "SUSE Package Hub"*).

2.8.1.3 etnaviv drivers for Vivante GPUs are available

The NXP* Layerscape* LS1028A/LS1018A System-on-Chip (SoC) contains a Vivante GC7000UL Graphics Processor Unit (GPU), and the NXP i.MX 8M SoC contains a Vivante GC7000L GPU.

As a technology preview, the SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) kernel includes `etnaviv`, a Display Rendering Infrastructure (DRI) driver for Vivante GPUs, and the `Mesa-dri` package contains a matching `etnaviv_dri` graphics driver library. Together they can avoid the need for third-party drivers and libraries.

Note

To use them, the Device Tree passed by the bootloader to the kernel needs to include a description of the Vivante GPU for the kernel driver to get loaded. You may need to contact your hardware vendor for a bootloader firmware upgrade.

2.8.1.4 lima driver for Arm Mali Utgard GPUs available

The Xilinx* Zynq* UltraScale*+ MPSoC contains an Arm* Mali*-400 Graphics Processor Unit (GPU).

Prior to SUSE Linux Enterprise Server for Arm 15 SP2, this GPU needed third-party drivers and libraries from your hardware vendor.

As a technology preview, the SUSE Linux Enterprise Server for Arm 15 SP2 kernel added `lima`, a Display Rendering Infrastructure (DRI) driver for Mali *Utgard* microarchitecture GPUs, such as Mali-400, and the `Mesa-dri` package contains a matching `lima_dri` graphics driver library.

> **Note**
>
> To use them, the Device Tree passed by the bootloader to the kernel needs to include a description of the Mali GPU for the kernel driver to get loaded. You may need to contact your hardware vendor for a bootloader firmware upgrade.

> **Note**
>
> The `panfrost` driver for Mali *Midgard* microarchitecture GPUs is supported since SUSE Linux Enterprise Server for Arm 15 SP2.

2.8.1.5 mali-dp driver for Arm Mali Display Processors available

The NXP* Layerscape* LS1028A/LS1018 System-on-Chip contains an Arm* Mali*-DP500 Display Processor.

As a technology preview, the SUSE Linux Enterprise Server for Arm 15 SP2 kernel added `mali-dp`, a Display Rendering Manager (DRM) driver for Mali Display Processors. It has undergone only limited testing because it requires an accompanying physical-layer driver for DisplayPort* output (see *Section 8.4.3, "No DisplayPort graphics output on NXP LS1028A and LS1018A"*).

2.8.1.6 Btrfs file system is enabled in U-Boot bootloader

For Raspberry Pi* devices, SUSE Linux Enterprise Server for Arm 12 SP3 and later include *Das U-Boot* as bootloader, in order to align the boot process with other platforms. By default, it loads GRUB as UEFI application from a FAT-formatted partition, and GRUB then loads Linux kernel and ramdisk from a file system such as Btrfs.

As a technology preview, SUSE Linux Enterprise Server for Arm 15 SP2 added a Btrfs driver to U-Boot for the Raspberry Pi (package `u-boot-rpiarm64`). This allows its commands `ls` and `load` to access files on Btrfs-formatted partitions on supported boot media, such as microSD and USB.

The U-Boot command `btrsubvol` lists Btrfs subvolumes.

2.8.2 Technology previews for Intel 64/AMD64 (x86-64)

2.8.2.1 Support for AMD Wheat Nas GPU

SLES 15 SP5 (prerelease) includes the kernel driver support for AMD Wheat Nas GPU (Navi32 dGPU). However because the corresponding firmware is still not publicly released yet, this feature is considered a technology preview.

3 Modules, extensions, and related products

This section comprises information about modules and extensions for SUSE Linux Enterprise Server 15 SP5 (prerelease). Modules and extensions add functionality to the system.

> **Note: Package and module changes in 15 SP5 (prerelease)**
>
> For more information about all package and module changes since the last version, see *Section 2.2.3, "Package and module changes in 15 SP5 (prerelease)"*.

3.1 Modules in the SLE 15 SP5 (prerelease) product line

The SLE 15 SP5 (prerelease) product line is made up of modules that contain software packages. Each module has a clearly defined scope. Modules differ in their life cycles and update timelines.

The modules available within the product line based on SUSE Linux Enterprise 15 SP5 (prerelease) at the release of SUSE Linux Enterprise Server 15 SP5 (prerelease) are listed in the *Modules and Extensions Quick Start* at https://susedoc.github.io/doc-sle/main/html/SLES-modules/ (draft version).

Not all SLE modules are available with a subscription for SUSE Linux Enterprise Server 15 SP5 (prerelease) itself (see the column *Available for*).

For information about the availability of individual packages within modules, see https://scc.suse.com/packages.

3.2 SLE extensions

SLE Extensions add extra functionality to the system and require their own registration key, usually at additional cost. Most extensions have their own release notes documents that are available from https://www.suse.com/releasenotes.

The following extensions are available for SUSE Linux Enterprise Server 15 SP5 (prerelease):

- SUSE Linux Enterprise Live Patching: https://www.suse.com/products/live-patching
- SUSE Linux Enterprise High Availability: https://www.suse.com/products/highavailability
- SUSE Linux Enterprise Workstation Extension: https://www.suse.com/products/workstation-extension

The following extension is not covered by SUSE support agreements, available at no additional cost and without an extra registration key:

- SUSE Package Hub: https://packagehub.suse.com/ (see *Section 5.11, "SUSE Package Hub"*)

3.3 Derived and related products

This sections lists derived and related products. Usually, these products have their own release notes documents that are available from https://www.suse.com/releasenotes.

- SUSE Linux Enterprise JeOS: https://www.suse.com/products/server/jeos (see *Section 4.3, "Minimal VM and Minimal Image"*)
- SUSE Linux Enterprise Desktop: https://www.suse.com/products/desktop
- SUSE Linux Enterprise Server for SAP Applications: https://www.suse.com/products/sles-for-sap
- SUSE Linux Enterprise for High-Performance Computing: https://www.suse.com/products/server/hpc
- SUSE Linux Enterprise Real Time: https://www.suse.com/products/realtime
- SUSE Manager: https://www.suse.com/products/suse-manager

4 Installation and upgrade

SUSE Linux Enterprise Server can be deployed in several ways:

- Physical machine
- Virtual host
- Virtual machine
- System containers
- Application containers

4.1 Installation

This section includes information related to the initial installation of SUSE Linux Enterprise Server 15 SP5 (prerelease).

Important: Installation documentation

The following release notes contain additional notes regarding the installation of SUSE Linux Enterprise Server. However, they do not document the installation procedure itself.

For installation documentation, see the *Deployment Guide* at https://susedoc.github.io/doc-sle/main/html/SLES-deployment/ (draft version).

4.1.1 New media layout

The set of media has changed with 15 SP2. There still are two different installation media, but the way they can be used has changed:

- You can install with registration using either the online-installation medium (as with SUSE Linux Enterprise Server 15 SP1) or the full medium.
- You can install without registration using the full medium. The installer has been added to the full medium and the full medium can now be used universally for all types of installations.
- You can install without registration using the online-installation medium. Point the installer at the required SLE repositories, combining the `install=` and `instsys=` boot parameters:
  - With the `install=` parameter, select a path that contains either just the product repository or the full content of the media.
  - With the `inst-sys=` parameter, point at the installer itself, that is, `/boot/ARCHITECTURE/root` on the medium.

  For more information about the parameters, see https://en.opensuse.org/SDB:Linuxrc#p_install.

4.1.2 SUSE Manager installation option not available

Because the release of version 4.4 of *SUSE Manager* was dropped, there is no *SUSE Manager Server 4.4* option available under the *Available Extensions and Modules* section during installation. See https://www.suse.com/releasenotes/x86_64/SUSE-MANAGER/4.3/index.html#_important_note for more information.

4.1.3 Storage requirements for Btrfs installation on PowerPC

There is a known issue with using guided partitioning to install SUSE Linux Enterprise Server on PowerPC architecture. The root filesystem needs more storage that the listed minimum. It is recommended to allocate at least 20 gigabytes.

4.1.4 Installation via SSH on s390x

Before starting the installation using `yast.ssh`, it is neccessary to set the environmental variable `QT_XCB_GL_INTEGRATION` to `xcb_egl`:

```
export QT_XCB_GL_INTEGRATION=xcb_egl
```

4.2 Upgrade-related notes

This section includes upgrade-related information for SUSE Linux Enterprise Server 15 SP5 (prerelease).

Important: Upgrade documentation

The following release notes contain additional notes regarding the upgrade of SUSE Linux Enterprise Server. However, they do not document the upgrade procedure itself.

For upgrade documentation, see the *Upgrade Guide* at https://susedoc.github.io/doc-sle/main/html/SLES-upgrade/ (draft version).

4.2.1 Hibernation requires manual intervention

Previously, it was possible for data loss to occur due to the system not hibernating correctly.

In 15 SP5 (prerelease), a sanity check was introduced to prevent this. It works by removing the kernel `resume` parameter if it points to a non-existent device. However, that means a system would not use the hibernation data. To fix it, do the following:

1. Edit `/etc/default/grub` and correct the `resume` parameter to point to an existing device.
2. Regenerate `initrd`.
3. Reboot.

4.2.2 Make sure the current system is up-to-date before upgrading

Upgrading the system is only supported from the most recent patch level. Make sure the latest system updates are installed by either running `zypper patch` or by starting the YaST module *Online Update*. An upgrade on a system that is not fully patched may fail.

4.2.3 Skipping service packs requires LTSS

Skipping service packs during an upgrade is only supported if you have a Long Term Service Pack Support contract. Otherwise, you need to first upgrade to SLE 15 SP4 before upgrading to SLE 15 SP5 (prerelease).

4.2.4 Direct upgrade from SLES 12 SP5 LTSS is not supported

You cannot upgrade directly from SUSE Linux Enterprise Server 12 SP5 LTSS to SUSE Linux Enterprise Server 15 15 SP5 (prerelease). SUSE Linux Enterprise Server 15 15 SP5 (prerelease) contains older package versions than SUSE Linux Enterprise Server 12 SP5 LTSS. Upgrading to SLES 15 15 SP5 (prerelease) causes package downgrades, which can break the system. To upgrade from SUSE Linux Enterprise Server 12 SP5 LTSS, you must upgrade directly to SUSE Linux Enterprise Server 15 SP6 or newer.

4.2.5 SMT has been replaced by RMT

SLE 12 is the last codestream that SMT (Subscription Management Tool) is available for.

When upgrading your OS installation to SLE 15, we recommend also upgrading from SMT to its replacement RMT (Repository Mirroring Tool). RMT provides the following functionality:

- Mirroring of SUSE-originated repositories for the SLE 12-based and SLE 15-based products your organization has valid subscriptions for.
- Synchronization of subscriptions from SUSE Customer Center using your organization's mirroring credentials. (These credentials can be found in SCC under *Select Organization*, *Organization*, *Organization Credentials*)
- Selecting repositories to be mirrored locally via `rmt-cli` tool.
- Registering systems directly to RMT to get required updates.
- Adding custom repositories from external sources and distributing them via RMT to target systems.
- Improved security with proxying: If you have strict security requirements, an RMT instance with direct Internet access can proxy to another RMT instance without direct Internet access.
- Nginx as Web server: The default Web server of RMT is Nginx which has a smaller memory footprint and comparable performance than that used for SMT.

Note that unlike SMT, RMT does not support installations of SLE 11 and earlier.

For more feature comparison between RMT and SMT, see https://github.com/SUSE/rmt/blob/master/docs/smt_and_rmt.md.

For more information about RMT, also see the new RMT Guide at https://documentation.suse.com/sles/html/SLES-all/book-rmt.html.

4.3 Minimal VM and Minimal Image

SUSE Linux Enterprise Server Minimal VM and Minimal Image is a slimmed-down form factor of SUSE Linux Enterprise Server that is ready to run in virtualization environments and the cloud. With SUSE Linux Enterprise Server Minimal VM and Minimal Image, you can choose the right-sized SUSE Linux Enterprise Server option to fit your needs.

SUSE provides virtual disk images for Minimal VM and Minimal Image in the file formats `.qcow2`, `.vhdx`, and `.vmdk`, compatible with KVM, Xen, OpenStack, Hyper-V, and VMware environments. All Minimal VM and Minimal Image images set up the same disk size (24 GB) for the system. Due to the properties of different file formats, the size of Minimal VM and Minimal Image image downloads differs between formats.

4.4 JeOS renamed Minimal VM and Minimal Image

We have received feedback from users confused by the name JeOS, as a matter of fact the acronym JeOS, which meant Just enough Operating System, was not well understood and could be confused with other images provided by SUSE or openSUSE.

We have decided to go with simplicity and rename JeOS by "Minimal VM" for all our Virtual Machine Images and "Minimal Image" for the Raspberry Pi Image. We have also removed a few other characters, in the full images name to make it more simple and clear:

- `SLES15-SP5-Minimal-VM.x86_64-kvm-and-xen-GM.qcow2`
- `SLES15-SP5-Minimal-VM.x86_64-OpenStack-Cloud-GM.qcow2`
- `SLES15-SP5-Minimal-VM.x86_64-MS-HyperV-GM.vhdx.xz`
- `SLES15-SP5-Minimal-VM.x86_64-VMware-GM.vmdk.xz`
- `SLES15-SP5-Minimal-VM.aarch64-kvm-GM.qcow2`
- `SLES15-SP5-Minimal-Image.aarch64-RaspberryPi-GM.raw.xz`

4.5 New Minimal VM and Minimal Image for IBM Z (s390x)

Per request, we have built and released new SLES Minimal VM and Minimal Image for IBM Z (s390x architecture).

Our Minimal VM and Minimal Image are supported on the 3 different type of storage used by IBM Z:

- DASDs (Direct Access Storage Devices) which works for z/VM guests and LPARs,
- FBA (Fixed-Block Architecture) which works for z/VM guests,
- and the regular qcow2 for KVM.

And to provide deployment choice, we have made 2 flavours for each type of storage, one with the regular `jeos-firstboot`, one with `cloud-init`.

So in total we have 6 new images:

KVM:

- `SLES15-SP5-Minimal-VM.s390x-kvm-GM.qcow2`
- `SLES15-SP5-Minimal-VM.s390x-Cloud-GM.qcow2`

DASD:

- `SLES15-SP5-Minimal-Image.s390x-dasd-GM.raw.xz`
- `SLES15-SP5-Minimal-Image.s390x-dasd-Cloud-GM.raw.xz`

FBA:

- `SLES15-SP5-Minimal-Image.s390x-fba-GM.raw.xz`
- `SLES15-SP5-Minimal-Image.s390x-fba-Cloud-GM.raw.xz`

Note that our regular images uses `jeos-firstboot` and Btrfs as the root filesystem by default, while our Cloud images use XFS as it’s usually recommended in cloud environment.

4.6 Generic Cloud image with cloud-init and jeos-firstboot fallback

SUSE has renamed the OpenStack-Cloud Minimal VM image to a generic Cloud image. The new filename is `SLES15-SP5-Minimal-VM.x86_64-Cloud-GM.qcow2`. This image is suitable for general cloud environments and local virtualization.

This image uses `cloud-init` as the primary first-boot configuration tool. If `cloud-init` does not detect a valid data source, the system falls back to `jeos-firstboot` for initialization. During this fallback, the system displays a clear message to notify you that `cloud-init` failed and `jeos-firstboot` is running.

This Cloud image has the following specific characteristics:

- SUSE provides this image only for the `x86_64` architecture.
- The image uses the XFS filesystem by default.
- The image does not contain Btrfs tools or the `firewalld` service.

4.6.1 Alternative modern Python 3 interpreter upgraded to Python 3.11

Warning: Changing system Python is not supported

Changing the system Python interpreter (located at `/usr/bin/python3`) is not supported. The recommended method of using non-system Python is an explicit shebang in the script file or using a virtual environment.

SLE 15 comes with a complete stack of Python 3.6 (interpreter, setuptools, wheel, pip plus hundreds of modules). Python 3.6 is fully supported by SUSE throught the *complete* lifecycle of SLE 15.

However, there are very few `pip` installable modules that are still compatible with 3.6 or older. Therefore, starting with SLE 15 SP4, SUSE introduced a new *Python 3* Module. It includes additional modern Python interpreter (plus `setuptool`, `wheel`, `pip` and pypi support, but no additional modules). This new allows more flexibility in providing a fully supported more recent Python interpreter in SLE.

Packages within the Python 3 module can be installed alongside existing Python packages and they can coexist within the same system without impacting any ongoing Python 3.6 workloads.

In SLE 15 SP5 the Python 3 Module ships with Python 3.11.

In SLE 15 SP4, which originally shipped with Python 3.10, Python 3.11 packages have been added in addition to the existing Python 3.10 packages. If you are using Python 3.10 in SLE 15 SP4 make sure to update the packages from the Python 3 Module to version 3.11 before upgrading to SLE SP5.

SLE 15 SP3 was shipped with the alternative Python version 3.9 included in *Basesystem* Module.

Python 3.11 is compatible with versions 3.10 and 3.9. Code written in 3.9 or 3.10 should run without or with only minimal changes in 3.11.

4.7 For more information

For more information, see *Section 5, "Changes affecting all architectures"* and the sections relating to your respective hardware architecture.

5 Changes affecting all architectures

Information in this section applies to all architectures supported by SUSE Linux Enterprise Server 15 SP5 (prerelease).

5.1 Basic utilities

5.1.1 `lzip` compressor available

The `lzip` compressor is now available in SUSE Linux Enterprise Server. This provides a command-line tool for compressing and decompressing files in the `.lz` format. It enables developers and projects to build packages from source files distributed in `.lz` format.

5.2 Containers

5.2.1 Podman upgrade from 3.4.x to 4.3.1

Podman 4.x is a major release with 60 new features and more than 50 bug fixes compared to Podman 3. It also includes a complete rewrite of the network stack.

Podman 4.x brings a new container network stack based on Netavark (https://github.com/containers/netavark) ↗, the new container network stack and Aardvark DNS server (https://github.com/containers/aardvark-dns) ↗ in addition to the existing container network interface (CNI) stack used by Podman 3.x . The new stack brings 3 important improvement:

- Better support for containers in multiple networks
- Better IPv6 support
- Better performance

To ensure that nothing break with this major change, the old CNI stack will remain the default on existing installations, while new installs will use Netavark.

New installations can opt to use CNI by explicitly specifying it via the `containers.conf` configuration file, using the `network_backend` field.

If you have run Podman 3.x before upgrading to Podman 4, Podman will continue to use CNI plugins as it had before. There is a marker in Podman's local storage that indicates this. In order to begin using Podman 4, you need to destroy that marker with `podman system reset`. This will destroy the marker, all of the images, all of the networks, and all of the containers.

Warning

Before testing Podman 4 and the new network stack, you will have to destroy all your current containers, images, and networks. You must export/save any import containers or images on a private registry, or make sure that your Dockerfiles are available for rebuilding and scripts/playbooks/states to reapply any settings, regenerate secrets, etc.

Last but not least CNI will be deprecated from upstream at a future date: https://github.com/containers/podman/tree/main/cni

For a complete overview of the changes, please check out the upstream 4.0.0 (https://github.com/containers/podman/releases/tag/v4.0.0) but also 4.1.1 (https://github.com/containers/podman/releases/tag/v4.1.1), 4.2.0 (https://github.com/containers/podman/releases/tag/v4.2.0) and 4.3.0 (https://github.com/containers/podman/releases/tag/v4.3.0) to be informed about all the new features and changes.

5.2.2 `suse/sle15` container uses NDB as the database back-end for RPM

Starting with SUSE Linux Enterprise 15 SP3, the `rpm` package in the `suse/sle15` container image no longer supports the BDB back-end (based on Berkeley DB) and switches to the NDB back-end. Tools for scanning, diffing, and building container image using the `rpm` binary of the host for introspection can fail or return incorrect results if the host's version of `rpm` does not recognize the NDB format.

To use such tools, make sure that the host supports reading NDB databases, such as hosts with SUSE Linux Enterprise 15 SP2 and later.

5.2.3 SLE BCI for kernel modules (Driver Toolkit)

A new Base Container Image (BCI) is now available to build and run kernel modules on SUSE Linux Enterprise Server, comparable to the upstream `driver-toolkit`. The container includes the necessary kernel headers for both default and RT kernels on `x86_64` and `aarch64` architectures, providing a simple way to prototype and test drivers without giving access to SLE kernel binaries.

5.2.4 Support for rootless Docker containers

Docker now supports running containers in rootless mode. You can now run Docker containers as a non-root user to increase system security. To run Docker in rootless mode, the `rootlesskit` package is required.

5.3 Databases

5.3.1 PostgreSQL 16 has been added

PostgreSQL 16 has been added to SUSE Linux Enterprise Server. For information about changes between PostgreSQL 16 and 15, see the upstream release notes (https://www.postgresql.org/docs/16/release.html).

At the same time, PostgreSQL 15 has been deprecated and has been moved to the Legacy module.

5.4 Desktop

5.4.1 nouveau disabled for Nvidia Turing and Ampere GPUs / openGPU recommendation

The `nouveau` driver is still considered experimental for Nvidia Turing and Ampere GPUs. Therefore it has been disabled by default on systems with these GPUs.

Instead of using the `nouveau` driver we recommend using Nvidia's new openGPU driver. Install this driver by installing these following packages:

- `nvidia-open-driver-G06-signed-kmp-default`
- `kernel-firmware-nvidia-gsp-G06`

Then uncomment the `options nvidia` line in the `/etc/modprobe.d/50-nvidia-default.conf` file so that it looks like the following afterwards:

```
### Enable support on *all* Turing/Ampere GPUs: Alpha Quality!
options nvidia NVreg_OpenRmEnableUnsupportedGpus=1
```

If you prefer using `nouveau` driver anyway, add `nouveau.force_probe=1` to your kernel boot parameters, and do not install the above openGPU package.

5.4.2 `gtk4-lang` Package Included for GTK 4 Localization

The `gtk4-lang` package is now included. This provides the translation files necessary for GTK 4 applications to display localized user interfaces.

5.5 Development

5.5.1 `python311-apache-libcloud` package added

SUSE Linux Enterprise Server now includes the `python311-apache-libcloud` package. This package provides the Apache Libcloud library for Python 3.11. You can use this package to run `salt-cloud` in Python 3.11 environments.

5.5.2 Support for Ansible interpreter

The Ansible packages (`ansible-core`) are included in SUSE Linux Enterprise Server 15 SP6 and newer. Support is limited to any Ansible playbooks or roles that are provided by SUSE as part of SUSE Linux Enterprise Server or SUSE Manager products, whether supplied directly (for example, via packages, ISO images, ready-to-run images) or dynamically generated by integrated tools within these products.

5.5.3 Alternative modern Python 3 interpreter upgraded to Python 3.11

Warning: Changing system Python is not supported

Changing the system Python interpreter (located at `/usr/bin/python3`) is not supported. The recommended method of using non-system Python is an explicit shebang in the script file or using a virtual environment.

SLE 15 comes with a complete stack of Python 3.6 (interpreter, setuptools, wheel, pip plus hundreds of modules). Python 3.6 is fully supported by SUSE throught the *complete* lifecycle of SLE 15.

However, there are very few `pip` installable modules that are still compatible with 3.6 or older. Therefore, starting with SLE 15 SP4, SUSE introduced a new *Python 3* Module. It includes additional modern Python interpreter (plus `setuptool`, `wheel`, `pip` and pypi support, but no additional modules). This new allows more flexibility in providing a fully supported more recent Python interpreter in SLE.

Packages within the Python 3 module can be installed alongside existing Python packages and they can coexist within the same system without impacting any ongoing Python 3.6 workloads.

In SLE 15 SP5 the Python 3 Module ships with Python 3.11.

In SLE 15 SP4, which originally shipped with Python 3.10, Python 3.11 packages have been added in addition to the existing Python 3.10 packages. If you are using Python 3.10 in SLE 15 SP4 make sure to update the packages from the Python 3 Module to version 3.11 before upgrading to SLE SP5.

SLE 15 SP3 was shipped with the alternative Python version 3.9 included in *Basesystem* Module.

Python 3.11 is compatible with versions 3.10 and 3.9. Code written in 3.9 or 3.10 should run without or with only minimal changes in 3.11.

5.5.3.1 Details

Although fully supported throughout the entire lifecycle of SLE 15, Python 3.6 is no longer developed by the Python community and it won’t any receive new functionality. Providing a modern Python versions as an alternative allows you to migrate to the newer version and to benefit from increased performance, compatibility with the latest syntax and new standard library versions. You will also be able to utilize contemporary modules provided by pypi.org.

The Python 3.11 interpreter and the SUSE provided Python modules will be supported in SLE 15 until at least 31st of December 2027. The interpreter will be regularly updated to the latest patch version. The Python modules will remain stable unless there is a security vulnerabiltiy or other significant issues. In such cases, the preferred approach will be to backport patches and maintain the released version.

- Added Python 3.11 interpreter and a basic set of modules (see the module's package list (https://scc.suse.com/packages?name=SUSE%20Linux%20Enterprise%20Server&version=15.4&arch=x86_64&query=python311&module=2405) for details)
  - All python 3.11 packages are prefixed with `python311-` to distinguish from the default (3.6) Python packages
  - Comes with significant performance improvments compared to Python 3.6
  - Compatible to Python 3.10 and 3.9
  - Supported on SLE 15 until at least 2027-12-31
- The Python 3 Module containing Python 3.11 will be activated by default
- The Python 3.6 interpreter and modules stay the default and remain supported throughout the SLE 15 lifecycle.
- We will continue delivering new interpreters (along with the respective `setuptools` / `wheel` / `pip` and pypi support, but no additional modules) with the Python 3 Module with each new service pack.

5.5.3.2 TCK compliance testing in SUSE Linux Enterprise

We run the TCK test suite provided by Oracle to ensure that our version of OpenJDK is in compliance with the Java specification.

5.5.4 Erlang upgraded to version 23

The Erlang runtime environment has been upgraded from version 22 to 23. This upgrade resolves a major regression in DTLS support that was introduced by the security fix for CVE-2022-37026.

Key improvements and potential incompatibilities in Erlang 23 include:

- **TLS 1.3:** Enabled by default as a supported TLS version.
- **DTLS:** Significantly improved DTLS support.
- **SSLv3 support removal:** Support for SSL 3.0 is completely removed.

- **Deprecated `erl_interface` parts removal:** The `erl_interface.h` header and all C functions with the `erl_` prefix are removed.
- **Deprecated `erlang:get_stacktrace/0` BIF:** Now returns an empty list instead of a stacktrace, scheduled for complete removal in Erlang 24.

Additionally, dependent packages `rabbitmq-server` and `elixir` have been rebuilt against Erlang 23 to ensure system stability.

5.5.5 Supported Java versions

The following Java implementations are available in SUSE Linux Enterprise Server 15 SP5 (prerelease):

Name (Package Name)	Version	Module	Support
OpenJDK (`java-11-openjdk`)	11	Base System	SUSE, L3, until 2026-12-31
OpenJDK (`java-17-openjdk`)	17	Base System	SUSE, L3, until 2028-06-30
OpenJDK (`java-1_8_0-openjdk`)	1.8.0	Legacy	SUSE, L3, until 2026-12-31
IBM Java (`java-1_8_0-ibm`)	1.8.0	Legacy	External, until 2025-04-30

5.6 Kernel

5.6.1 Kernel limits

This table summarizes the various limits which exist in our recent kernels and utilities (if related) for SUSE Linux Enterprise Server 15 SP5 (prerelease).

SLES 15 SP5 (prerelease) (Linux 5.14.21)	AMD64/Intel 64 (x86_64)	IBM Z (s390x)	POWER (ppc64le)	ARMv8 (AArch64)
CPU bits	64	64	64	64

SLES 15 SP5 (prerelease) (Linux 5.14.21)	AMD64/Intel 64 (x86_64)	IBM Z (s390x)	POWER (ppc64le)	ARMv8 (AArch64)
Maximum number of logical CPUs	8192	256	2048	768
Maximum amount of RAM (theoretical/certified)	>1 PiB/64 TiB	10 TiB/ 256 GiB	1 PiB/64 TiB	256 TiB/n.a.
Maximum amount of user space/kernel space	128 TiB/ 128 TiB	n.a.	4 PiB[1]/2 EiB	256 TiB/ 256 TiB
Maximum amount of swap space	Up to 29 * 64 GB	Up to 30 * 64 GB		
Maximum number of processes	1,048,576			
Maximum number of threads per process	Upper limit depends on memory and other parameters (tested with more than 120,000)[2].			
Maximum size per block device	Up to 8 EiB on all 64-bit architectures			
FD_SETSIZE	1024			

[1] By default, the user space memory limit on the POWER architecture is 128 TiB. However, you can explicitly request mmaps up to 4 PiB.

[2] The total number of all processes and all threads on a system may not be higher than the "maximum number of processes".

5.6.2 Restoring default Btrfs file compression

Previously in kernel 5.14, it was possible to disable compression by passing an empty string instead of explicitly mentioning `none` or `no`.

In SLES 15 SP5 (prerelease), this behavior is changed to the more expected one. From kernel 5.14 onwards, empty string will reset the default setting instead of disabling compression.

5.6.3 Support AMD Instinct™ MI300A

In SLES 15 SP5 (prerelease), we support RAS, edac and mce modules for AMD Instinct™ MI300A. RAS errors can now be detected by mce_amd module and decoding, and logging will be handled by the amd64_edac modules.

5.7 Miscellaneous

- support firmware updates from LVFS using UEFI capsule update features

5.7.1 Apache's Portable Runtime Library devel package renaming

These two packages have been renamed:

- `libapr1-devel` to `apr-devel`
- `libapr-util1-devel` to `apr-util-devel`

5.7.2 `suseconnect -r` failure in Secure Boot mode

When system is booted in Secure Boot mode, due to outdated `qclib` and `s390-tools` packages, `suseconnect -r` fails with the following error:

```
ExecuteError: Cmd: [read_values -s], RC: 1,
Error: exit status 1,
Output: Error: Unable to open configuration, return_code =-2
```

To fix the issue, run the system in non-Secure Boot mode for the update to the latest SP, then update `s390-tools` and `qclib` to the latest version. Deregister the system and register it again in Secure Boot mode.

5.7.3 New systems-management module

A new module called *Systems Management Module* has been added. This module will include systems-management packages, such as Ansible.

5.7.4 TigerVNC's `vncserver` has been removed

The `vncserver` wrapper script around Xvnc has been removed upstream.

See the official documentation (https://github.com/TigerVNC/tigervnc/blob/04f19667cd53be-caab97d888c3cfb7246338a238/unix/vncserver/HOWTO.md) for information on using Xvnc in version 1.11.0 or newer.

5.7.5 `cpuid` has been added

The `cpuid` package has been added. It provides detailed information about the CPU. For more information see http://etallen.com/cpuid.html.

5.7.6 Apache HTTP Server upgraded to version 2.4.66

The Apache HTTP Server (`apache2`) has been upgraded from version 2.4.51 to 2.4.66. This update introduces the following security and behavioral changes:

- The default value of `LimitRequestBody` has changed from unlimited (`0`) to `1GB`. Uploads exceeding `1GB` now fail with an HTTP `413` error. To restore unlimited uploads, configure `LimitRequestBody 0`.
- You must use `AddHandler` instead of `AddType` to connect requests to handlers (CVE-2024-38476).
- Unsafe rewrite rules that use backreferences as the first path segment now fail or return an HTTP `404` error (CVE-2024-38475, CVE-2024-38474). Configure the rewrite flags `UnsafePrefixStat` or `UnsafeAllow3F` to restore previous behavior.
- CGI scripts can no longer set the `Content-Length` or `Transfer-Encoding` headers (CVE-2024-24795). To restore the `Content-Length` header, set the environment variable `ap_trust_cgilike_cl` to a non-empty value.
- Support for opportunistic TLS upgrades (`SSLEngine optional`) has been removed (CVE-2025-49812).
- The duplicate keys `BusyWorkers` and `IdleWorkers` have been removed from `mod_status`, and `GracefulWorkers` has been added.
- The default MIME type for `.js` and `.mjs` files has changed from `application/javascript` to `text/javascript`.

5.8 Networking

5.8.1 `ddclient` updated to version 3.10.0

The `ddclient` package has been updated from version 3.8.0 to version 3.10.0. This update restores support for the Cloudflare dynamic DNS protocol. In older versions, `ddclient` failed to update dynamic IP addresses via Cloudflare. The new version also removes the test dependency on the unavailable `perl(HTTP::Message::PSGI)` module.

5.8.2 `frr` (FRRouting Routing daemon) has been added

The `frr` package has been added. It manages TCP/IP based routing protocols.

FRR is a fork of Quagga, which has stopped development in 2018. Its developers moved to the FRR project and thus Quagga will receive no further updates.

We recommend migrating to `frr`. The configuration is mostly backward compatible, including the `vtysh` shell to configure the routing protocols. However, there were several changes, improvements, and new functionality added to `frr`.

See https://frrouting.org/ for more information.

5.8.3 Samba

The version of Samba shipped with SUSE Linux Enterprise Server 15 SP5 (prerelease) delivers integration with Windows Active Directory domains. In addition, we provide the clustered version of Samba as part of SUSE Linux Enterprise High Availability 15 SP5 (prerelease).

5.8.4 NFS

5.8.4.1 NFSv4

NFSv4 with IPv6 is only supported for the client side. An NFSv4 server with IPv6 is not supported.

5.9 Security

5.9.1 ClamAV has been updated to version 1.0

The `clamav` package has been updated to version 1.0. This major version update introduces several new features, security enhancements, and bug fixes:

- Expanded decryption support, including decrypted Microsoft Excel (XLS) files.
- Support for scanning images embedded in HTML emails.
- Performance optimizations and improved parsing of various file formats.
- Internal modernizations, including further integration of the Rust-based components for memory-safe execution.

5.9.2 strongSwan has been updated to version 5.9.11

The `strongswan` package has been updated to version 5.9.11. This update resolves a USGv6 compliance test failure for IPsec IKEv2 connections where key derivation would previously fail for non-HMAC based algorithms (such as AES-XCBC). Additionally, this update removes an unstable custom patch (Marvell patch) that previously caused daemon instability and random crashes. Numerous other backported patches have been replaced by upstream fixes included in this release, improving overall security and stability.

5.9.3 TLS 1.1 and 1.0 are no longer recommended for use

The TLS 1.0 and 1.1 standards have been superseded by TLS 1.2 and TLS 1.3. TLS 1.2 has been available for considerable time now.

SUSE Linux Enterprise Server packages using OpenSSL, GnuTLS, or Mozilla NSS already support TLS 1.3. We recommend no longer using TLS 1.0 and TLS 1.1, as SUSE plans to disable these protocols in a future service pack. However, not all packages, for example, Python, are TLS 1.3-enabled yet as this is an ongoing process.

5.9.4 New 4096-bit signing key

SUSE Linux Enterprise 15 product family switched over from a RSA 2048-bit signing key to a new RSA 4096-bit key. This change covers RPM packages, package repositories and ISO signatures. For more information see https://documentation.suse.com/sles/15-SP5/html/SLES-all/cha-update-preparation.html#sec-update-preparation-update

5.9.5 Replacement of gpg as recommended tool for file encryption

A new `rage-encryption` package has been added. The package provides the `rage` executable. `rage` is an implementation of the `age` file encryption format using Rust.

This replaces `gpg` as the preferred tool over `gpg` for file encryption. The main reasons are that `gpg` has a history of security issues, and is well known for its complexity. In comparison, `rage` focuses on usability and security. It especially tries to make key handling as simple as handling SSH keys (essentially short lines of text), which it also supports.

For more information, see:

- https://github.com/str4d/rage
- https://age-encryption.org/v1

5.9.6 GRUB 2: TPM measurement support integrated by default

By default, the GRUB 2 bootloader now integrates Trusted Platform Module (TPM) measurement support directly into the standard `grub.efi` binary. This integration eliminates the need for a secondary `grub-tpm.efi` binary and simplifies bootloader maintenance. On SUSE Linux Enterprise, TPM measurement is not enabled by default to prevent stability issues on specific hardware. To enable TPM measurement, set `TRUSTED_BOOT="yes"` in `/etc/sysconfig/bootloader`.

5.9.7 haveged switch-root service removed

The `haveged-switch-root.service` has been removed from SUSE Linux Enterprise Server 15 SP4, SP5, and SP6. Implementing a robust switch-root service for `haveged` is difficult and was prone to security and synchronization issues. Since Linux kernel 5.4, a HAVEGED-inspired entropy gathering algorithm has been included directly in the kernel, making `haveged-switch-root.service` unnecessary.

To resolve any remaining early-boot entropy requirements safely, `haveged-once.service` has been introduced instead, which runs with the `--once` flag during early boot and exits cleanly before switching root.

5.10 Storage and file systems

5.10.1 Deprecation of legacy XFS v4 filesystem format

SUSE Linux Enterprise maintains backward compatibility for legacy XFS v4 filesystems via the kernel configuration `CONFIG_XFS_SUPPORT_V4=y`. This support ensures uninterrupted access for existing environments. However, XFS v4 is deprecated, and creating new v4 filesystems is strongly discouraged.

September 2030 is the EOL target for upstream Linux kernel support for XFS v4. Any release post-September 2030 based on updated upstream kernels will omit XFS v4 support entirely. Legacy v4 filesystems will no longer be mountable on those versions.

Upgrading underlying filesystems from v4 to v5 is transparent to applications and requires no changes to userspace software.

To check the format version of an existing XFS filesystem, run:

```
xfs_info /path/to/mountpoint
```

An output including `crc=1` indicates the modern v5 format, while `crc=0` indicates the legacy v4 format.

There is no in-place or online upgrade path from XFS v4 to v5. Migration requires backing up data, reformatting the block device to XFS v5, and restoring the content. Migrating the root filesystem and any other critical data partitions requires planned downtime.

The operating system generates warnings when v4 format interaction occurs:

- `mkfs.xfs` displays a warning if forced to format as v4 (`crc=0`):

  ```
  V4 filesystems are deprecated and will not be supported by future versions
  ```

- The kernel logs a notification upon mounting a v4 filesystem (printed once per boot cycle, upon first mount of a v4 filesystem):

  ```
  Deprecated V4 format (crc=0) will not be supported after September 2030.
  ```

5.10.2 Booting from NVMe-oF™ over TCP

SUSE Linux Enterprise Server 15 SP5 (prerelease) and later support booting from NVMe-oF™ over TCP according to the NVM Express® Boot Specification 1.0 (https://nvmexpress.org/wp-content/uploads/NVM-Express-Boot-Specification-2022.11.15-Ratified.pdf). Previous versions of SLE supported boot from NVMe-oF over Fibre Channel only.

For more information see the documentation (https://documentation.suse.com/sles/15-SP5/html/SLES-all/cha-nvmeof.html#sec-nvmeof-boot).

5.10.3 dracut default persistent policy change

The previously used `by-path` policy had the following shortcomings:

- doesn’t work for multi-path
- very fragile for iSCSI
- can change with hardware changes

To avoid the above problems, the persistent policy of `dracut` is now set to `by-uuid`.

5.10.4 Comparison of supported file systems

SUSE Linux Enterprise was the first enterprise Linux distribution to support journaling file systems and logical volume managers in 2000. Later, we introduced XFS to Linux, which allows for reliable large-scale file systems, systems with heavy load, and multiple parallel reading and writing operations. With SUSE Linux Enterprise 12, we started using the copy-on-write file system Btrfs as the default for the operating system, to support system snapshots and rollback.

The following table lists the file systems supported by SUSE Linux Enterprise.

Support status: + supported / – unsupported

Feature	Btrfs	XFS	Ext4	OCFS 2[1]
Supported in product	SLE	SLE	SLE	SLE HA
Data/metadata journaling	N/A[2]	– / +	+ / +	– / +
Journal internal/external	N/A[2]	+ / +	+ / +	+ / –

Feature	Btrfs	XFS	Ext4	OCFS 2[1]
Journal checksumming	N/A[2]	+	+	+
Subvolumes	+	–	–	–
Offline extend/shrink	+ / +	– / –	+ / +	+ / –[3]
Inode allocation map	B-tree	B+-tree	Table	B-tree
Sparse files	+	+	+	+
Tail packing	–	–	–	–
Small files stored inline	+ (in metadata)	–	+ (in inode)	+ (in inode)
Defragmentation	+	+	+	–
Extended file attributes/ACLs	+ / +	+ / +	+ / +	+ / +
User/group quotas	– / –	+ / +	+ / +	+ / +
Project quotas	–	+	+	–
Subvolume quotas	+	N/A	N/A	N/A
Data dump/restore	–	+	–	–
Block size default	4 KiB[4]			
Maximum file system size	16 EiB	8 EiB	1 EiB	4 PiB
Maximum file size	16 EiB	8 EiB	1 EiB	4 PiB

[1] OCFS 2 is fully supported as part of the SUSE Linux Enterprise High Availability.

[2] Btrfs is a copy-on-write file system. Instead of journaling changes before writing them in-place, it writes them to a new location and then links the new location in. Until the last write, the changes are not "committed". Because of the nature of the file system, quotas are implemented based on subvolumes (`qgroups`).

[3] To extend an OCFS 2 file system, the cluster must be online but the file system itself must be unmounted.

[4] The block size default varies with different host architectures. 64 KiB is used on POWER, 4 KiB on other systems except for AArch64, which offers both 4K and 64K. The actual size used can be checked with the command `getconf PAGE_SIZE`.

Additional notes

Maximum file size above can be larger than the file system's actual size because of the use of sparse blocks. All standard file systems on SUSE Linux Enterprise Server have LFS, which gives a maximum file size of 2^{63} bytes in theory.

The numbers in the table above assume that the file systems are using a 4 KiB block size which is the most common standard. When using different block sizes, the results are different.

In this document:

- 1024 Bytes = 1 KiB
- 1024 KiB = 1 MiB;
- 1024 MiB = 1 GiB
- 1024 GiB = 1 TiB
- 1024 TiB = 1 PiB
- 1024 PiB = 1 EiB.

See also http://physics.nist.gov/cuu/Units/binary.html.

Some file system features are available in SUSE Linux Enterprise Server 15 SP5 (prerelease) but are not supported by SUSE. By default, the file system drivers in SUSE Linux Enterprise Server 15 SP5 (prerelease) will refuse mounting file systems that use unsupported features (in particular, in read-write mode). To enable unsupported features, set the module parameter `allow_unsupported=1` in `/etc/modprobe.d` or write the value `1` to `/sys/module/MODULE_NAME/parameters/allow_unsupported`. However, note that setting this option will render your kernel and thus your system unsupported.

5.10.5 Supported Btrfs features

The following table lists supported and unsupported Btrfs features across multiple SLES versions.

Support status: + supported / – unsupported

Feature	SLES 11 SP4	SLES 12 SP5	SLES 15 GA	SLES 15 SP1	SLES 15 SP2	SLES 15 SP3
Copy on write	+	+	+	+	+	+
Free space tree (Free Space Cache v2)	–	–	–	+	+	+
Snapshots/subvolumes	+	+	+	+	+	+
Swap files	–	–	–	+	+	+
Metadata integrity	+	+	+	+	+	+
Data integrity	+	+	+	+	+	+
Online metadata scrubbing	+	+	+	+	+	+
Automatic defragmentation	–	–	–	–	–	–
Manual defragmentation	+	+	+	+	+	+
In-band deduplication	–	–	–	–	–	–
Out-of-band deduplication	+	+	+	+	+	+
Quota groups	+	+	+	+	+	+
Metadata duplication	+	+	+	+	+	+
Changing metadata UUID	–	–	–	+	+	+

Feature	**SLES 11 SP4**	**SLES 12 SP5**	**SLES 15 GA**	**SLES 15 SP1**	**SLES 15 SP2**	**SLES 15 SP3**
Multiple devices	–	+	+	+	+	+
RAID 0	–	+	+	+	+	+
RAID 1	–	+	+	+	+	+
RAID 5	–	–	–	–	–	–
RAID 6	–	–	–	–	–	–
RAID 10	–	+	+	+	+	+
Hot add/remove	–	+	+	+	+	+
Device replace	–	–	–	–	–	–
Seeding devices	–	–	–	–	–	–
Compression	–	+	+	+	+	+
Big metadata blocks	–	+	+	+	+	+
Skinny metadata	–	+	+	+	+	+
Send without file data	–	+	+	+	+	+
Send/receive	–	+	+	+	+	+
Inode cache	–	–	–	–	–	–
Fallocate with hole punch	–	+	+	+	+	+

5.11 SUSE Package Hub

SUSE Package Hub brings open-source software packages from openSUSE to SUSE Linux Enterprise Server and SUSE Linux Enterprise Desktop.

Usage of software from SUSE Package Hub is not covered by SUSE support agreements. At the same time, usage of software from SUSE Package Hub does not affect the support status of your SUSE Linux Enterprise systems. SUSE Package Hub is available at no additional cost and without an extra registration key.

5.11.1 Important package additions to SUSE Package Hub

Among others, the following packages have been added to SUSE Package Hub:

5.12 System management

5.12.1 Salt package `python311-salt` added

The new `python311-salt` package (based on Python 3.11) is now available on SLE 15 SP5 (prerelease). This package is co-installable with the existing `python3-salt` package (based on Python 3.6). Both packages will be supported simultaneously for a transition period of roughly six months to allow a smooth migration to the newer Python 3.11-based package. By default, the active Salt binaries can be configured using `update-alternatives`. To install the dependencies for `python311-salt`, make sure that the Python 3 Module is enabled on your system.

5.12.2 Effective user limits in systemd setup

Before, the lookup of the effective session limit in a systemd setup was not trivial. Now these new properties have been added:

- `EffectiveMemoryMax`
- `EffectiveMemoryHigh`
- `EffectiveTasksMax`

5.12.3 Silence `KillMode=None` messages

The log level of the deprecation warnings regarding `killmode=None` have been reduced. Instead of `warning`, they are now logged at the `debug` log level.

5.12.4 `yast2-iscsi-client` drops `open-iscsi` and `iscsiuio` as dependencies

The `yast2-iscsi-client` package no longer automatically installs `open-iscsi` and `iscsiuio`. The two packages need to be installed manually before using `yast2-iscsi-client`.

5.12.5 Support for System V init.d scripts is deprecated

systemd in SUSE Linux Enterprise Server 15 SP5 (prerelease) automatically converts System V `init.d` scripts to service files. Support for System V init.d scripts is deprecated and will be removed with the next major version of SUSE Linux Enterprise Server. In the next major version of SUSE Linux Enterprise Server, systemd will also stop converting System V `init.d` scripts to systemd service files.

To prepare for this change, use the automatically generated systemd service files directly instead of using System V `init.d` scripts. To do so, copy the generated service files to `/etc/systemd/system`. To then control the associated services, use `systemctl`.

The automatic conversion provided by systemd (specifically, `systemd-sysv-generator`) is only meant to ensure backward compatibility with System V init.d scripts. To take full advantage of systemd features, it can be beneficial to manually rewrite the service files.

This deprecation also causes the following changes:

- The `/etc/init.d/halt.local` initscript is deprecated. Use systemd service files instead.
- `rc`*`SERVICE`* controls of systemd services are deprecated. Use systemd service files instead.
- `insserv.conf` is deprecated.

5.12.6 Searching packages across all SLE modules

In SLE 15 SP5 (prerelease) you can search for packages both within and outside of currently enabled SLE modules using the following command:

```
zypper search-packages -d SEARCH_TERM
```

This command contacts the SCC and searches all modules for matching packages. This functionality makes it easier for administrators and system architects to find the software packages needed.

5.12.7 Performance Co-Pilot (PCP) upgraded to version 6.2.0 to resolve vulnerabilities

Previous versions of Performance Co-Pilot (PCP) had security vulnerabilities that allowed local users to escalate privileges. To address these issues, the PCP package has been upgraded to version 6.2.0. This upgrade resolves multiple security vulnerabilities, including CVE-2023-6917, CVE-2024-45769, and CVE-2024-45770.

5.13 Virtualization

For more information about acronyms used below, see https://documentation.suse.com/sles/15-SP5/html/SLES-all/book-virtualization.html.

> **Important: Virtualization limits and supported hosts/guests**
>
> These release notes only document changes in virtualization support compared to the immediate previous service pack of SUSE Linux Enterprise Server. Full information regarding virtualization limits for KVM and Xen as well as supported guest and host systems is now available as part of the SUSE Linux Enterprise Server documentation.
>
> See the *Virtualization Guide* at https://susedoc.github.io/doc-sle/main/html/SLES-virtualization/cha-virt-support.html (draft version).

5.13.1 KVM

We increase the support from 288 up 768 VCPU per virtual Machine.

5.13.2 Xen

Xen has been updated to version 4.17:

- The x86 MCE command line option info is now updated.
- `__ro_after_init` support, for marking data as immutable after boot.

- The project has officially adopted 4 directives and 24 rules of MISRA-C, added MISRA-C checker build integration, and defined how to document deviations.
- IOMMU superpage support on x86, affecting PV guests as well as HVM/PVH ones when they don't share page tables with the CPU (HAP/EPT/NPT).
- Support for `VIRT_SSBD` and `MSR_SPEC_CTRL` for HVM guests on AMD.
- Improved TSC, CPU, and APIC clock frequency calibration on x86.
- Support for Xen using x86 Control Flow Enforcement technology for its own protection. Both Shadow Stacks (ROP protection) and Indirect Branch Tracking (COP/JOP protection).
- Add `mwait-idle` support for SPR and ADL on x86.
- Extend security support for hosts to 12 TiB of memory on x86.
- Add command line option to set cpuid parameters for dom0 at boot time on x86.
- It is possible to use PV drivers with dom0less guests, allowing statically booted dom0less guests with PV devices.
- Add Xue - console over USB 3 Debug Capability.
- dropped support for the (x86-only) `vesa-mtrr` and `vesa-remap` command line options
- `launch_security`: Use SEV-ES policy=0x07 if host supports it

5.13.3 QEMU

QEMU has been updated to version 7.1:

- Raise the maximum number of vCPUs a VM can have to 1024 (only 768 is supported)
- Improve dependency handling (for example, what is recommended compared to what is required)
- Add the `qemu-headless` subpackage that brings in all the packages that are needed for creating VMs with tools like `virt-install` or `VirtManager`; it can be run either locally or from a remote host
- The old `qemu-binfmt` wrappers around the various `qemu-$ARCH` Linux user emulation binaries are not necessary any longer
- Enable` aio=io_uring` on all KVM architectures

For more information see the following:

- https://wiki.qemu.org/ChangeLog/7.1
- https://wiki.qemu.org/ChangeLog/7.0
- https://qemu-project.gitlab.io/qemu/about/removed-features.html
- https://qemu-project.gitlab.io/qemu/about/deprecated.html

Note: Deprecation notice

In previous versions, if no explicit image format was provided, some QEMU tools tried to guess the format of the image, and then process it accordingly. Because this feature is a potential source of security issues, it has been deprecated and removed. It is now necessary to explicitly specify the image format. For more information, see https://qemu-project.gitlab.io/qemu/about/removed-features.html#qemu-img-backing-file-without-format-removed-in-6-1 .

5.13.4 libvirt

`libvirt` has been updated to version 9.0.0:

- New `virt-qemu-sev-validate` utility for validating the measurement reported for a domain launched with AMD SEV
- New subpackage `libvirt-client-qemu` providing client utilities to interact with QEMU-specific features of `libvirt`
- Migration to `/usr/etc` : saving user changed configuration files in `/etc` and restoring them while an RPM update

For more information see the following:

- https://libvirt.org/news.html#v9-0-0-2023-01-16
- https://libvirt.org/news.html#v8-10-0-2022-12-01
- https://libvirt.org/news.html#v8-9-0-2022-11-01
- https://libvirt.org/news.html#v8-8-0-2022-10-03
- https://libvirt.org/news.html#v8-7-0-2022-09-01

5.13.5 VMware

5.13.5.1 open-vm-tools

`open-vm-tools` has been updated to version 12.1.5:

- Migration of PAM settings to `/usr/lib/pam.d`
- Remove libgrpc++, libgrpc, and libprotobuf from containerinfo `Requires` section. The dependencies will be added automatically.
- Add `containerInfo` plugin
- Add dependencies on grpc, protobuf, and containerd for container introspection
- A number of Coverity reported issues have been addressed.
- The `deployPkg` plugin may prematurely reboot the guest VM before cloud-init has completed user data setup. If both the Perl based Linux customization script and cloud-init run when the guest VM boots, the deployPkg plugin may reboot the guest before cloud-init has finished. The `deployPkg` plugin has been updated to wait for a running cloud-init process to finish before the guest VM reboot is initiated.
- A `SIGSEGV` may be encountered when a non-quiesing snapshot times out
- Unwanted `vmtoolsd` service error message if not on a VMware hypervisor When `open-vm-tools` comes preinstalled in a base Linux release, the `vmtoolsd` services are started automatically at system start and desktop login. If running on physical hardware or in a non-VMware hypervisor, the services will emit an error message to the systemd's logging service before stopping.

5.13.6 Others

5.13.6.1 Deprecating host network management with `libvirt`

Managing host network devices with `libvirt`'s `virInterface*` APIs is deprecated and may be removed in a future service pack. Any users of these APIs or `virsh`'s `iface-*` subcommands should use the distribution networking tools instead. For more information see the *Basic networking* chapter of the *Administration Guide* at https://documentation.suse.com/sles/15-SP5/single-html/SLES-administration/#cha-network.

5.13.6.2 NVIDIA GRID

Support for NVIDIA Virtual GPU (vGPU) v15.1 has been added. The support does NOT include NVIDIA vGPU live migration support.

5.13.6.3 `virt-manager`

`virt-manager` has been updated to version 4.1.0:

- Specifying `--boot` no longer implies `no_install=yes`
- add UI and cli support for `qemu-vdagent` channel
- cli: More `--iothreads` suboptions
- cli: Add support for URL query with disks

5.13.6.4 `sanlock`

`sanlock` has been updated to version 3.8.5:

- Add support for Python 3
- Add support for 4k sector size
- Support `SANLOCK_RUN_DIR` and `SANLOCK_PRIVILEGED` environment variables

5.13.6.5 Perl-Sys-Virt

Update to 0.9.0: * Add all new APIs and constants in libvirt 9.0.0

5.13.6.6 `numactl`

`numactl` has been updated to version 2.0.15.0.g01a39cb:

- Update to support multiple nodes
- numademo: Add a new test for multiple-preferred-nodes policy
- numactl: Simplify preferred selection

5.13.6.7 libguestfs

`libguestfs` has been updated to version 1.48.4:

- Drop reiserfs
- Multiple fixes to the OCaml bindings
- Inspection of guests which use LUKS encryption on top of LVM logical volumes should now work `guestfs_remove_drive` has been deprecated and now returns an error.
- `guestfs_add_drive` no longer supports hotplugging
- In `guestfs_xfs_admin` the `lazycounter` parameter is deprecated because it is no longer supported in recent versions of XFS.
- The User-mode Linux ("uml") backend has been removed.
- This release has moved many virt tools like virt-builder, virt-cat, virt-customize, virt-df, etc. to the guestfs-tools project. This makes libguestfs a bit easier to build and manage.
- We now use the `qemu`/`libvirt` feature `-cpu max` to select the best CPU to run the appliance.
- The `qemu -enable-fips` option is no longer used. It was not needed and has been deprecated by `qemu`.
- Using the equivalent SeaBIOS feature instead of `qemu's Serial Graphics Adapter option ROM
- Renamed packages:
  - `guestfs-winsupport` → `libguestfs-winsupport`
  - `guestfsd` → `libguestfsd`

- New packages:
  - `libguestfs`, `libguestfs-typelib-Guestfs`,
  - `libguestfs-gobject`, `libguestfs-gobject-devel`
  - `libguestfs-rescue`, `libguestfs-rsync`, `libguestfs-xfs`
- Dropped packages:
  - `libguestfs-test`

5.13.6.8 virt-v2v

Update to version 2.0.7:

- Virt-v2v has been modularised allowing external programs to examine the state of the conversion and inject their own copying step. Further enhancements will be made to this new architecture in forthcoming releases.
- The command line is almost identical apart from some debugging features that were removed (see below). The only significant difference is that the output format (-of) now has to be specified if it is different from the input format, whereas previous versions of virt-v2v would use the same output formatas input format automatically.
- A lot of time was spent improving the performance of virt-v2v in common cases.
- Many bug fixes and performance enhancements were made to oVirt imageio output (Nir Soffer).
- There is a new virt-v2v-in-place(1) tool which replaces the existing virt-v2v --in-place option.
- Virt-v2v can now convert guests which use LUKS encrypted logical volumes.
- Option -oo rhv-direct has been replaced by -oo rhv-proxy, and direct mode (which is much faster) is now the default when writing to oVirt, with proxy mode available for restricted network configurations.
- The following command line options were removed: -print-estimate, --debug-overlays, --no-copy.
- Virt-v2v no longer installs the RHEV-APT tool in Windows guests. This tool was deprecated and then removed in oVirt 4.3.
- Deprecated tool virt-v2v-copy-to-local has been removed.

5.13.6.9 `sevctl`

The `sevctl` package version 0.3.2 has been added. It replaces the deprecated `sev-tool` package.

5.13.6.10 Kubevirt support

We provide L3 support for KubeVirt in SLES 15 SP5 (prerelease). It is supported for the lifecycle of the product only if you are using the latest version update (N+1) or version N available. KubeVirt is not supported in via LTSS / Extended.

If you'd like to report a bug with KubeVirt, use the following components in Bugzilla: `KVM stack: QEMU/KVM, libvirt, virt-manager, kubevirt`.

5.13.6.11 KubeVirt support for AMD SEV and SEV-ES

We now provide AMD Secure Encrypted Virtualization (SEV) and Secure Encrypted Virtualization-Encrypted State (SEV-ES) support for KubeVirt. This support allows you to secure guest virtual machines with memory encryption. You can now run encrypted container-native virtualization workloads.

6 POWER-specific changes (ppc64le)

Information in this section applies to SUSE Linux Enterprise Server for POWER 15 SP5 (prerelease).

Also see the following notes elsewhere:

- *Section 4.1.3, “Storage requirements for Btrfs installation on PowerPC”*

6.1 Security

There were also the following changes:

- POWER10 performance enhancements for cryptography: NSS FreeBL

6.1.1 POWER10 performance enhancements for cryptography: nettle

There were the following improvements:

- Use defined structure constants of P1305 in `asm.m4`
- Implement Poly1305 single block update based on radix 2^64
- Workaround for qemu bug affecting the ppc instruction vmsumudm

6.1.2 POWER10 performance enhancements for cryptography: libgcrypt

There were the following improvements:

- Chacha20/poly1305 - Optimized chacha20/poly1305 for P10 operation
- AES-GCM: Bulk implementation of AES-GCM acceleration for ppc64le
- hwf-ppc: fix missing HWF_PPC_ARCH_3_10 in HW feature

6.1.3 POWER10 performance enhancements for cryptography: OpenSSL

There were the following improvements:

- AES-GCM
  - AES-GCM performance optimization with stitched method for p9+ ppc64le
- Chacha20

- chacha20 performance optimizations for ppc64le with 8x lanes, Performance increase around 50%

6.2 Storage

6.2.1 NVMe-oF on LVM with multiple NVMe disks

The installation of NVMe Over Fabrics (NVMe-oF) does not work with Logical Volume Management when the root Volume Group utilizes a storage pool consisting of multiple NVMe-oF disks, which necessitates discovery beyond the boot disk.

6.3 Virtualization

There were also the following changes:

- Added NVMf-FC kdump support

6.3.1 Support for the new H_WATCHDOG hypercall included with P10 firmware

Added a `pseries-wdt` driver that exposes these hypercall-based watchdog timers to userspace via the Linux watchdog API.

6.3.2 `ibmveth` driver performance improvement

Changed the `ibmveth` driver to implement a more efficient way of giving packets to the hypervisor for transmit. Instead of DMA mapping and unmapping every outgoing data buffer, the buffer is copied into a reusable DMA mapped buffer. Also implemented multiple transmit queues for parallel packet processing.

6.3.3 Adding changes in `ibmvnic` driver to assign IRQ affinity to all

CPUs after hotplug events (CPU or vnic device): irqbalance daemon provides `--banmod` option so that IRQ affinities will be preserved with driver settings. But the irqbalance `--banmod` option is not working with vnic IRQs and this feature fixes the bug in the irqbalance daemon code.

6.4 Miscellaneous

6.4.1 Transactional memory is deprecated and disabled

On POWER9, transactional memory is partially emulated by the hypervisor, but this does not give the expected performance.

Therefore, transactional memory is now disabled by default in the kernel. For legacy applications on platforms that still support transactional memory, it can be enabled with the `ppc_tm=on` kernel parameter.

7 IBM Z-specific changes (s390x)

Information in this section applies to SUSE Linux Enterprise Server for IBM Z and LinuxONE 15 SP5 (pre-release). For more information, see https://www.ibm.com/docs/en/linux-on-systems?topic=distributions-suse-linux-enterprise-server ↗

7.1 Hardware

There were the following hardware-related changes:

- Support for IBM z16 and LinuxONE 4 in qclib
- Exploitation support of new IBM Z crypto hardware in zcrypt DD
- Support for a new set of crypto performance counters of the IBM z16 and LinuxONE 4
- Add new CPU-MF Counters for IBM z16 and LinuxONE 4 in kernel,` s390-tools` and libpfm part.
- Support for "Crypto Compliance" feature of the IBM z16 and LinuxONE 4 Processor-Activity-Instrumentation Facility in kernel and` s390-tools`

7.2 Networking

7.2.1 Enablement for MIO Instructions - kernel and `rdma-core` parts

Make use of the new PCI Load/Store instructions in the `rdma-core` package for increased performance.

7.3 Performance

7.3.1 zlib CRC32 optimization for s390x

This new feature utilizes SIMD instructions to accelerate the zlib CRC32 implementation, resulting in significant performance improvements for applications that rely heavily on zlib.

7.4 Security

7.4.1 Secure Execution

If you want to use IBM Secure Execution see https://www.ibm.com/docs/en/linux-on-systems?topic=execution-prerequisites-restrictions

7.4.2 openCryptoki: `p11sak` support Dilithium and Kyber keys

The openCryptoki `p11sak` tool has been extended to manage Dilithium and Kyber keys.

7.4.3 In-kernel crypto: SIMD implementation of `chacha20`

Recent kernel releases provide support for the `chacha20` cipher and the goal of this feature is to use z System SIMD instruction to accelerate the `chacha20` implementation on IBM Z and LinuxONE.

7.4.4 openCryptoki ep11 token: master key consistency

Ensuring that all APQNs used by an openCryptoki ep11 token are configured with the same master key, if not print an error message and fail initialization.

7.4.5 zcrypt DD: Exploitation Support of new IBM Z Crypto Hardware for kernel and `s390-tools`

zcrypt DD: exploitation support of new IBM Z crypto hardware by recognizing and supporting the CEX8 adapters.

7.4.6 Display PAI (Processor Activity Instrumentation) CPACF counters (`s390-tools`)

Provides a tool to display CPACF usage counters from the IBM z16 and LinuxONE 4 Processor Activity Instrumentation Facility.

7.4.7 openCryptoki EP11 token: IBM z16 and LinuxONE 4 support

The openCryptoki EP11 token supports new vendor specific mechanisms for quantum safe ciphers (Dilithium and Kyber) supported by the CEX8S crypto adapter.

7.4.8 openCryptoki EP11 token: vendor-specific key derivation

Support of a new function from EP11 7.2 by the openCryptoki EP11 token. Comprises the support of vendor-specific mechanisms for a key derivation function used with cryptocurrencies and Schnorr signatures.

7.4.9 openCryptoki: support crypto profiles

Support for a new configuration option to restrict cryptographic functions/mechanisms.

7.4.10 openCryptoki key generation with expected MKVP only on CCA and EP11 tokens

For the EP11 and CCA tokens allow to configure expected master key verification pattern (MKVPs) configured in the crypto adapter(s) and upon generation of a new secure keys ensures that the MKVP of the generated key is equal to an expected MKVP.

7.4.11 libica: extend statistics to reflect security measures (crypto)

Adds new `-k` parameter to `icastats` to display detailed counters to reflect security measures (key size/curve/hash size).

7.4.12 `libica`: eliminate SW fall backs - stage 2

Eliminates implementations of SW fallback functions for RSA in `libica`.

7.4.13 `zcryptctl` support for control domains - kernel and s390-tools parts

Improves access control to crypto resources via device nodes, for example, for Docker containers by allowing to assign control domains to a device node created by `zcryptctl`.

7.4.14 openCryptoki: PKCS #11 3.1 - support CKA_DERIVE_TEMPLATE

In openCryptoki, support the new attribute `CKA_DERIVE_TEMPLATE` introduced with PKCS #11 v3.1.

7.4.15 `p11-kit`: add IBM specific mechanisms and attributes

Adds support for IBM-specific attributes and mechanisms to the PKCS11 client-server implementation of `p11-kit`.

7.4.16 `libica`: FIPS 140-3 compliance

Update of `libica` to fulfill FIPS 140-3 requirements.

7.5 Storage

7.5.1 Transparent DASD PPRC (Peer-to-Peer Remote Copy) handling

Enables user of Linux on Z to use DASD volumes in a PPRC (Peer-to-Peer Remote Copy) relation like a normal DASD volume thereby enabling platform support along with improving user experience.

7.5.2 zdev: Site-aware device configuration

Enables a Linux on Z admin to prepare a Linux root/boot volume so that it can be started on a different system (site) or with different device IDs and parameters without manually changing the configuration.

7.5.3 `zipl` support for Secure Boot IPL and Dump from ECKD DASD

Enables `zipl` to use an ECKD DASD for Secure Boot IPL and Dump.

7.5.4 `zipl`: Site-aware environment block

Enables a Linux on Z admin to create a `zipl` configuration file that enables different kernel parameters depending on the IPL site.

7.5.5 Transparent PCI device recovery

Improves reliability and reduces downtime by using cooperative recovery strategies that allow the drivers to recover from error scenarios automatically (without intervention from userspace), and without going through a complete tear-down and subsequent re-init.

7.5.6 ROCE: Independent Usage of Secondary Physical Function

This feature enables zLinux running in an LPAR to use the Phisical Function (PF) that corresponds to the second oprt of a ConnectX-5/6 card even if the PF that corresponds to the first port is assigned to a different LPAR.

7.6 Virtualization

The following new features are supported in SUSE Linux Enterprise Server 15 SP5 (prerelease) under KVM:

7.6.1 KVM: Enable GISA support for Secure Execution guests

GISA (Guest Interruption State Area) is now supported for Secure Execution Guests. This allows the direct injection of interrupts into running VMs, which results in reduced processing overhead.

7.6.2 Enhanced Interpretation for PCI Functions

For improving performance, the interpretive execution of the PCI store and PCI load instructions are enabled. Further improvement is achieved by enabling the Adapter-Event-Notification Interpretation which enables customers to run PCI I/O intensive workloads in KVM guests. This feature also allows KVM guests to use the Shared Memory Communications - Direct (SMC-D) protocol.

7.6.3 `vfio-ap` enhancements

This feature adds hotplug support, which allows to dynamically assign and remove crypto adapters to or from a running KVM guest. This additionally provides a tool to persistently define the configuration of crypto adapters and domains that are allowed for passthrough usage as well as the configuration of the matrix/vfio-ap devices to be passed through to avoid that customers have to reconfigure after every IPL.

7.6.4 Secure Execution guest dump encryption with customer keys, tool to process encrypted Secure Execution guest dumps

These features implements dumps created in a way that can only be decrypted by the owner of the guest image and be used for problem determination hence mitigating the outcomes of situations where `kdump` hypervisor-initiated dumps are not helpful.

7.6.5 KVM: Attestation support for Secure Execution (crypto), KVM: Secure Execution Attestation Userspace Tool

Provides attestation support, for example, for external frameworks, specific deployment models or potentially regulatory requirements.

7.6.6 KVM: Provide virtual CPU topology to guests

This feature allows to specify a virtual CPU topology which can help to configure the guest for better performance.

7.6.7 KVM: Allow long kernel command lines for Secure Execution guests and QEMU

This feature allows Secure Execution guests to use larger command lines.

7.6.8 Allow to list persisted `driverctl` definitions

Updated `driverctl` to version newer than 0.111 because the previous version did not allow to list persisted override definitions.

7.6.9 KVM: Enablement of device busid for subchannels

Displaying the original CCW device numbers for` vfio-ap` devices improves the usability of `vfio-ccw` device passthrough to KVM guests.

7.7 Kernel

7.7.1 NVMe stand-alone dump support

Supports the HW roadmap, full exploitation of NVMe on the IBM Z platform.

7.7.2 Support IBM z16 and LinuxONE 4 Processor-Activity-Instrumentation Facility

Support for the Crypto Compliance feature of the IBM z16 and LinuxONE 4 Processor-Activity-Instrumentation Facility.

7.7.3 New CPU-MF Counters for IBM z16 and LinuxONE 4

Added new CPU-MF Counters for IBM z16 and LinuxONE 4.

7.7.4 Support Processor Activity Instrumentation Extension 1

Support for a new set of crypto performance counters of the IBM z16 and LinuxONE 4.

7.7.5 Add additional information to SCLP CPI

Includes distribution, release number, and detailed kernel version in CPI data to be displayed on the HMC.

7.8 Miscellaneous

7.8.1 Auto scale `crashkernel` size with hardware changes

`kdump` now has a configuration option called `KDUMP_AUTO_RESIZE`. This option makes the boot-time init script call `kdumptool-calibrate --shrink`. This performs the same reservation size estimate that `kdumptool calibrate` normally does and shrinks the memory reservation to the calculated value by writing to `/sys/kernel/kexec_crash_size`.

YaST now has an option to turn this run-time auto-detection on. The result will be a very large `crashkernel=` value (around half the RAM) passed to the kernel and the reservation is reduced during boot using the `KDUMP_AUTO_RESIZE` option.

7.8.2 Enabled installer to configure PCI-attached networking devices

The installer now supports PCI-attached networking devices.

7.8.3 Removed ESCON from installer

Removed the ESCON entry from the menu because it is no longer supported by the YaST module for network configuration.

7.8.4 Workaround for missing password prompt on fully encrypted disks during bare metal installation

During a bare metal installation on IBM Z (s390x) with a fully encrypted disk, the password prompt to decrypt the disk may not be visible in the HMC terminal. This is because systemd directs the password prompt to Plymouth, which does not display messages on line-based consoles.

To work around this issue, add the `plymouth.enable=0` parameter to the boot command line of the installation media. This disables Plymouth and ensures the password prompt is correctly displayed in the HMC terminal.

8 Arm 64-bit-specific changes (AArch64)

Information in this section applies to SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease).

8.1 System-on-Chip driver enablement

SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) includes driver enablement for the following System-on-Chip (SoC) chipsets:

- AMD* Opteron* A1100
- Ampere* X-Gene*, eMAG*, Altra*, *Altra Max*, ***AmpereOne****
- AWS* Graviton, Graviton2, Graviton3
- Broadcom* BCM2837/BCM2710, BCM2711
- Fujitsu* A64FX
- Huawei* Kunpeng* 916, Kunpeng 920
- Marvell* ThunderX*, ThunderX2*; OCTEON TX*; Armada* 7040, Armada 8040
- NVIDIA* **Grace**; Tegra* X1, Tegra X2, Xavier*, **Orin**; BlueField*, *BlueField-2*
- NXP* i.MX 8M, 8M Mini; Layerscape* LS1012A, LS1027A/LS1017A, LS1028A/LS1018A, LS1043A, LS1046A, LS1088A, LS2080A/LS2040A, LS2088A, LX2160A
- Qualcomm* Centriq* 2400
- Rockchip RK3399
- Socionext* SynQuacer* SC2A11
- Xilinx* Zynq* UltraScale*+ MPSoC

Note

Driver enablement is done as far as available and requested. Refer to the following sections for any known limitations.

Some systems might need additional drivers for external chips, such as a Power Management Integrated Chip (PMIC), which may differ between systems with the same SoC chipset.

For booting, systems need to fulfill either the Server Base Boot Requirements (SBBR) or the Embedded Base Boot Requirements (EBBR), that is, the Unified Extensible Firmware Interface (UEFI) either implementing the Advanced Configuration and Power Interface (ACPI) or providing a Flat Device Tree (FDT) table. If both are implemented, the kernel will default to the Device Tree; the kernel command line argument `acpi=force` can override this default behavior.

Check for SUSE *YES!* certified systems, which have undergone compatibility testing.

8.2 New features

8.2.1 Driver enablement for NVIDIA Orin

SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) adds initial enablement for the NVIDIA Orin* SoC (T234), which is found on Jetson* AGX Orin, Jetson Orin NX and Jetson Orin Nano System-on-Modules (SoM).

Warning: Firmware/kernel ABI break in NVIDIA JetPack 6.0 SDK

NVIDIA JetPack* 6.0 boot firmware and Linux kernel 6.5 changed the Application Binary Interface (ABI) for numbering General Purpose Input/Output (GPIO) pins — specifically the main GPIO ports X, Y, Z, AC, AD, AE, AF and AG — referenced in the machine-specific vendor Device Tree (DT) binary for NVIDIA Orin based systems.

A SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) kernel maintenance update adopts the behavior of the latest kernels and is *fully* compatible only with JetPack 6.0 and later:

Firmware	SLES 15 SP5 (prerelease) GM	SLES 15 SP5 (prerelease) QU1	SLES 15 SP5 (prerelease) QU2	SLES 15 SP5 (prerelease) QU3
JetPack 5.x	+	+	+	–
JetPack 6.x	–	–	–	+

This is known to affect some peripherals on the following modules or systems:

- NVIDIA Jetson AGX Orin Developer Kit:
  - USB Type-C controller
  - Ethernet controller

    - Camera sensors
    - PCIe endpoint mode
- NVIDIA Jetson Orin NX:
    - Camera sensors
- NVIDIA Jetson Orin Nano:
    - Camera sensors
- For others, check with your system vendor.

For installed systems running JetPack firmware versions prior to 6.0, suggested steps are:

1. Update the kernel packages from the installed system, but do not reboot into the new kernel yet.
2. Shut the system down in order to enter its Recovery Mode.
3. Flash the latest JetPack firmware according to system vendor instructions. Be careful to not overwrite your SUSE Linux Enterprise Server for Arm installation. For example: `sudo ./flash.sh device-identifier-and-boot-medium external`
4. Boot the system into the new kernel.

For new installations, consider these options:

- SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) Quarterly Update 3 (QU3) media from SUSE Customer Center are expected to include the updated kernel suited for systems with NVIDIA JetPack 6.0 and later. Flash the latest vendor firmware before installing from these updated media.
- In the meantime, with 15 SP5 (prerelease) Quarterly Update 2 and earlier media, consider installing from an earlier version of JetPack and following above update steps.
- If a firmware downgrade is not available for your system, contact SUSE Customer Support for whether a custom kISO medium can be created.

SUSE Linux Enterprise Server for Arm 15 SP6 will only support NVIDIA JetPack 6.0 and later.

Drivers for the integrated, NVIDIA *Ampere* microarchitecture-based Graphics Processor Unit (GPU) are not included (*Section 8.4.2, “No graphics drivers on NVIDIA Jetson”*).

8.2.2 Driver enablement for NVIDIA Grace

A SUSE Linux Enterprise Server for Arm 15 SP4 maintenance update added partial enablement for the NVIDIA Grace* SoC (T241).

SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) adds further enablement for the NVIDIA Grace SoC.

This also covers the NVIDIA Grace Hopper* SoC with an integrated, *Hopper* microarchitecture-based Graphics Processor Unit (GPU). Drivers for this integrated GPU are not included (*Section 8.4.1, “No graphics drivers on NVIDIA Grace Hopper”*).

NVIDIA recommend to use `kernel-64kb` on NVIDIA Grace and Grace Hopper SoCs (*Section 8.3, “64K page size kernel flavor is supported”*).

8.3 64K page size kernel flavor is supported

SUSE Linux Enterprise Server for Arm 12 SP2 and later kernels have used a page size of 4K. This offers the widest compatibility also for small systems with little RAM, allowing to use Transparent Huge Pages (THP) where large pages make sense.

As a technology preview, SUSE Linux Enterprise Server for Arm 15 SP3 added a kernel flavor `64kb`, offering a page size of 64 KiB and physical/virtual address size of 52 bits. Same as the `default` kernel flavor, it does not use preemption.

SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) largely removes this technology preview status, offering support for `kernel-64kb` on select platforms, such as NVIDIA Grace* (*Section 8.2.2, “Driver enablement for NVIDIA Grace”*). KVM virtualization remains a technology preview on this `64kb` kernel flavor (*Section 2.8.1.1, “KVM virtualization with 64K page size kernel flavor”*).

Note: Default file system no longer needs to be changed

SUSE Linux Enterprise Server for Arm 15 SP4 and later allow the use of Btrfs based file systems with 4 KiB block size also with 64 KiB page size kernels.

Important: Swap needs to be re-initialized

After booting the 64K kernel, any swap partitions need to re-initialized to be usable. To do this, run the `swapon` command with the `--fixpgsz` parameter on the swap partition. Note that this process deletes data present in the swap partition (for example, suspend data). In this example, the swap partition is on `/dev/sdc1`:

```
swapon --fixpgsz /dev/sdc1
```

Warning: RAID 5 uses page size as stripe size

It is currently possible to configure stripe size by setting the following kernel parameter:

```
echo 16384 > /sys/block/md1/md/stripe_size
```

Keep in mind that `stripe_size` must be in multiples of 4KB and not bigger than `PAGE_SIZE`. Also, it is only supported on systems where `PAGE_SIZE` is not 4096, such as arm64.

Avoid RAID 5 volumes when benchmarking 64K vs. 4K page size kernels.

See the *Storage Guide* for more information on software RAID.

Note: Cross-architecture compatibility considerations

The SUSE Linux Enterprise Server 15 SP5 (prerelease) kernels on x86-64 use 4K page size.

The SUSE Linux Enterprise Server for POWER 15 SP5 (prerelease) kernel uses 64K page size.

8.3.1 Hyper-V packages are available for AArch64 architecture

We have added the `hyper-v` package to the AArch64 software channels. Previously, this package was only available in public cloud images on Microsoft Azure. You can now install and use the standard Hyper-V integration tools on all supported AArch64 virtual machines.

8.4 Known limitations

8.4.1 No graphics drivers on NVIDIA Grace Hopper

The NVIDIA Grace Hopper* System-on-Chip contains an integrated, *Hopper* microarchitecture-based Graphics Processor Unit (GPU).

SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) maintenance updates currently provide packages `nvidia-open-driver-G06-signed-kmp-default` and `kernel-firmware-nvidia-gspx-G06` in version `535.104.05`, which does not yet enable NVIDIA Grace Hopper GH200.

Check for maintenance updates of those packages with version `545.29.02` or later, or contact the system vendor or chip vendor NVIDIA for whether third-party graphics drivers are available for SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease).

Note: PCIe GPUs not affected

Discrete GPU cards with *Hopper* microarchitecture, such as NVIDIA H100, are already enabled in shipping package versions.

8.4.2 No graphics drivers on NVIDIA Jetson

The NVIDIA* Tegra* System-on-Chip chipsets include an integrated Graphics Processor Unit (GPU).

SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease) does not include graphics drivers for any of the NVIDIA Jetson*, NVIDIA IGX or NVIDIA DRIVE* platforms.

Contact the chip vendor NVIDIA for whether third-party graphics drivers are available for SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease).

8.4.3 No DisplayPort graphics output on NXP LS1028A and LS1018A

The NXP* Layerscape* LS1028A/LS1018A System-on-Chip contains an Arm* Mali*-DP500 Display Processor, whose output is connected to a DisplayPort* TX Controller (HDP-TX) based on Cadence* High Definition (HD) Display Intellectual Property (IP).

A Display Rendering Manager (DRM) driver for the Arm Mali-DP500 Display Processor is available as technology preview (*Section 2.8.1.5, “mali-dp driver for Arm Mali Display Processors available”*).

However, there was no HDP-TX physical-layer (PHY) controller driver ready yet. Therefore no graphics output will be available, for example, on the DisplayPort* connector of the NXP LS1028A Reference Design Board (RDB).

Contact the chip vendor NXP for whether third-party graphics drivers are available for SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease).

Alternatively, contact your hardware vendor for whether a bootloader update is available that implements graphics output, allowing to instead use `efifb` framebuffer graphics in SUSE Linux Enterprise Server for Arm 15 SP5 (prerelease).

Note

The Vivante GC7000UL GPU driver (`etnaviv`) is available as a technology preview (*Section 2.8.1.3, “etnaviv drivers for Vivante GPUs are available”*).

9 Removed and deprecated features and packages

This section lists features and packages that were removed from SUSE Linux Enterprise Server or will be removed in upcoming versions.

Note: Package and module changes in 15 SP5 (prerelease)

For more information about all package and module changes since the last version, see *Section 2.2.3, “Package and module changes in 15 SP5 (prerelease)”*.

9.1 Removed features and packages

The following features and packages have been removed in this release.

- The `haveged-switch-root.service` has been removed. For more information, see *Section 5.9.7, “*haveged *switch-root service removed”*.
- `libmfx`
- The `samba-ad-dc-libs` package has been removed. It was previously available as technical preview.

- Setting up Kerberos with LDAP backend via YaST has been removed.
- The thunderbolt-user-space package has been removed.
- `zypper-docker` has been removed.
- `xpram` device driver has been removed.
- PostgreSQL 13, previously moved to the Legacy module, has been removed.

Also see the following notes elsewhere:

- *Section 5.7.4, "TigerVNC's `vncserver` has been removed"*

9.2 Deprecated features and packages

The following features and packages are deprecated and will be removed in a future version of SUSE Linux Enterprise Server.

- Legacy XFS v4 filesystems are deprecated. For details, see *Section 5.10.1, "Deprecation of legacy XFS v4 filesystem format"*.
- The standalone `wire` package is deprecated and will not receive further updates. Grafana now vendors the `wire` package internally.
- `docker-runc` has been deprecated and will be removed in SLES 15 SP6. Use `runc` instead.
- PHP 7.4 will be removed in SLES 15 SP7.
- TLS 1.0 and 1.1 are deprecated and will be removed in a future service pack of SUSE Linux Enterprise Server 15. For more information, see *Section 5.9.3, "TLS 1.1 and 1.0 are no longer recommended for use"*.
- OSN support on IBM Z has been deprecated.
- The `mkinitrd` wrapper has been replaced with `dracut` everywhere and will be removed in the next major version of SUSE Linux Enterprise Server.
- The `lftp_wrapper` package has been deprecated and will be removed in the near future. It is still available as an `update-alternative` for `ftp`, but it is no longed used by default. The default implementation of `ftp` is now the `lftp` executable.

- Support for System V `init.d` scripts is deprecated and will be removed with the next major version of SUSE Linux Enterprise Server. In consequence, the `/etc/init.d/halt.local` initscript, `rcSERVICE` controls, and `insserv.conf` are also deprecated. For more information, see *Section 5.12.5, "Support for System V init.d scripts is deprecated"*.

- `lftp_wrapper` is deprecated. Use `lftp` directly instead.

- On the POWER architecture, transactional memory is deprecated. For more information, see *Section 6.4.1, "Transactional memory is deprecated and disabled"*.

- The `opa-fmgui` package is not maintained upstream anymore. It has been deprecated, moved to the Legacy module, and will be removed in a future service pack.

- NIS is deprecated and will be removed with the next major version of SUSE Linux Enterprise Server. This includes packages implementing NIS, like `ypserv`. NIS code will be removed from SUSE tools and all NIS client code will be dropped with the next major version of SUSE Linux Enterprise Server.

- 389 Directory Server replaces OpenLDAP as the LDAP directory service. OpenLDAP has been re-added to SLE 15 SP5 and SLE SP6 to prolong the time window for a migration to 389 Directory Server. Support for OpenLDAP on SLE 15 will end with the SLE 15 SP6 lifecycle. Future versions and service packs of SLE will not include OpenLDAP.

9.2.1 Public Cloud module deprecations

The following packages in the Public Cloud module have been deprecated and will be removed in SUSE Linux Enterprise Server 15 SP6:

- `azure-cli-acr`
- `azure-cli-acs`
- `azure-cli-advisor`
- `azure-cli-ams`
- `azure-cli-appservice`
- `azure-cli-backup`
- `azure-cli-batch`

- azure-cli-batchai
- azure-cli-billing
- azure-cli-cdn
- azure-cli-cloud
- azure-cli-cognitiveservices
- azure-cli-component
- azure-cli-configure
- azure-cli-consumption
- azure-cli-container
- azure-cli-cosmosdb
- azure-cli-dla
- azure-cli-dls
- azure-cli-dms
- azure-cli-eventgrid
- azure-cli-eventhubs
- azure-cli-extension
- azure-cli-feedback
- azure-cli-find
- azure-cli-interactive
- azure-cli-iot
- azure-cli-keyvault
- azure-cli-lab
- azure-cli-monitor
- azure-cli-network

- azure-cli-profile
- azure-cli-rdbms
- azure-cli-redis
- azure-cli-reservations
- azure-cli-resource
- azure-cli-role
- azure-cli-search
- azure-cli-servicebus
- azure-cli-servicefabric
- azure-cli-sql
- azure-cli-storage
- azure-cli-taskhelp
- azure-cli-vm
- blue-horizon-config-deploy-cap-eks
- cfn-lint
- gcimagebundle
- google-cloud-sdk
- google-compute-engine
- google-daemon
- google-startup-scripts
- python-azure-storage
- python-ravello-sdk
- python-vsts-cd-manager
- python-vsts

- `regionServiceClientConfigHP`
- `regionServiceClientConfigSAPAzure`
- `regionServiceClientConfigSAPEC2`
- `regionServiceClientConfigSAPGCE`
- `terraform-provider-aws`
- `terraform-provider-azurerm`
- `terraform-provider-external`
- `terraform-provider-google`
- `terraform-provider-helm`
- `terraform-provider-kubernetes`
- `terraform-provider-local`
- `terraform-provider-null`
- `terraform-provider-openstack`
- `terraform-provider-random`
- `terraform-provider-susepubliccloud`
- `terraform-provider-template`
- `terraform-provider-tls`
- `WALinuxAgent`

9.2.2 Miscellaneous

- `sev-tool` has been deprecated. Use `sevctl` instead.
- `gnote` has been deprecated. Use `bijiben` instead.
- We have switched from `openmpi2` to `openmpi4` as the default `openmpi` implementation. This is because `openmpi2` and `openmpi3` have been EOL for some time now. They will be removed in SLES 15 SP6.

10 Obtaining source code

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A Changelog for 15 SP5 (prerelease)

A.1 Pre-release

A.1.1 New

- Added note about the new Cloud image with fallback configuration to *Section 4.6, "Generic Cloud image with cloud-init and jeos-firstboot fallback"* (Jira (https://jira.suse.com/browse/PED-3404) ↗).
- Added note about the addition of the `python311-apache-libcloud` package (*Section 5.5.1, "`python311-apache-libcloud` package added"*, Jira (https://jira.suse.com/browse/PED-14451) ↗, Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1257268) ↗).
- Added note about rootless Docker support to *Section 5.2.4, "Support for rootless Docker containers"* (Jira (https://jira.suse.com/browse/PED-6484) ↗).
- Added note about upgrading Performance Co-Pilot (PCP) to version 6.2.0 (*Section 5.12.7, "Performance Co-Pilot (PCP) upgraded to version 6.2.0 to resolve vulnerabilities"*, Jira (https://jira.suse.com/browse/PED-8349) ↗, Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1217826) ↗).
- Added note about direct upgrade from SLES 12 SP5 LTSS to SLES 15 15 SP5 (prerelease) being unsupported (*Section 4.2.4, "Direct upgrade from SLES 12 SP5 LTSS is not supported"*, Jira (https://jira.suse.com/browse/PED-7994) ↗).
- Added note about the `ddclient` update to version 3.10.0 (*Section 5.8.1, "`ddclient` updated to version 3.10.0"*, Jira (https://jira.suse.com/browse/PED-7810) ↗, Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1213181) ↗).
- Added note about Hyper-V guest support for AArch64 to *Section 8.3.1, "Hyper-V packages are available for AArch64 architecture"* (Jira (https://jira.suse.com/browse/PED-2918) ↗).
- Uncommented and updated *Appendix B, Kernel parameter changes* with default settings changes for 15 SP5 (prerelease) (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1218234) ↗, Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1219295) ↗).

- Upgraded Apache HTTP Server to 2.4.66 to address several security vulnerabilities and describe breaking configuration changes (*Section 5.7.6, "Apache HTTP Server upgraded to version 2.4.66"*, Jira (https://jira.suse.com/browse/PED-16335) ↗).

- Added note about AMD SEV and SEV-ES support for KubeVirt to *Section 5.13.6.11, "KubeVirt support for AMD SEV and SEV-ES"* (Jira (https://jira.suse.com/browse/SLE-23244) ↗).

- Added link to the Public Cloud Information Tracker (PINT) to *Section 2.5, "Documentation and other information"* (Jira (https://jira.suse.com/browse/DOCTEAM-351) ↗).

- Added note about addition of PostgreSQL 16 to *Section 5.3.1, "PostgreSQL 16 has been added"* (Jira (https://jira.suse.com/browse/PED-5594) ↗).

- Added release note for the Erlang 23 upgrade (*Section 5.5.4, "Erlang upgraded to version 23"*, Jira (https://jira.suse.com/browse/PED-6272) ↗, Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1207113) ↗).

- Added note about `haveged-switch-root.service` removal and replacement to *Section 5.9.7, "`haveged` switch-root service removed"* (Jira (https://jira.suse.com/browse/PED-6185) ↗).

- Added note about the availability of the lzip compressor (*Section 5.1.1, "`lzip` compressor available"*, Jira (https://jira.suse.com/browse/PED-5910) ↗).

- Added note about SLE BCI for kernel modules to *Section 5.2.3, "SLE BCI for kernel modules (Driver Toolkit)"* (Jira (https://jira.suse.com/browse/PED-5576) ↗).

- Added note about strongSwan 5.9.11 update to *Section 5.9.2, "strongSwan has been updated to version 5.9.11"* (Jira (https://jira.suse.com/browse/PED-4590) ↗).

- Added legacy XFS v4 format deprecation notice to *Section 9, "Removed and deprecated features and packages"* and *Section 5.10.1, "Deprecation of legacy XFS v4 filesystem format"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1275537) ↗, Jira (https://jira.suse.com/browse/DOCTEAM-2174) ↗).

- Added note about `python311-salt` package to *Section 5.12.1, "Salt package `python311-salt` added"* (Jira (https://jira.suse.com/browse/PED-13284) ↗).

- Added note about ClamAV 1.0 update to *Section 5.9.1, "ClamAV has been updated to version 1.0"* (Jira (https://jira.suse.com/browse/PED-4597) ↗).

- Added note about GRUB 2 TPM measurement default support to *Section 5.9.6, "GRUB 2: TPM measurement support integrated by default"* (Jira (https://jira.suse.com/browse/PED-126) ↗).

- Added note about gtk4-lang inclusion to *Section 5.4, "Desktop"* (Jira (https://jira.suse.com/browse/PED-14448)).
- Added `wire` package deprecation notice to *Section 9, "Removed and deprecated features and packages"* (Jira (https://jira.suse.com/browse/PED-15902)).
- Added workaround for missing password prompt on fully encrypted disks during bare metal installation on IBM Z (*Section 7.8.4, "Workaround for missing password prompt on fully encrypted disks during bare metal installation"*, Jira (https://jira.suse.com/browse/PED-11267))

A.1.2 Updated

- Updated tracking reference for NIS deprecation (Jira (https://jira.suse.com/browse/PED-3865)).

A.2 2026-07-09

A.2.1 New

- Added note about PostgreSQL 13 being removed to *Section 9, "Removed and deprecated features and packages"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1217483))

A.3 2026-02-27

A.3.1 New

- Added `libmfx` to *Section 9.1, "Removed features and packages"* (Jira (https://jira.suse.com/browse/PED-10574))

A.3.2 Updated

- Changed footnote in *Section 5.10.4, "Comparison of supported file systems"* to mention that AArch64 offers both 4K and 64K block size (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1212186))

A.4 2025-10-31

A.4.1 New

- Added note about removal of `docker-runc` to *Section 9, "Removed and deprecated features and packages"* (Jira (https://jira.suse.com/browse/PED-4018))
- Mention firmware updates from LVFS using UEFI capsule update features in *Section 5.7, "Miscellaneous"* (Jira (https://jira.suse.com/browse/PED-229))

- *Section 5.10.2, "Booting from NVMe-oF™ over TCP"* (Jira (https://jira.suse.com/browse/PED-2966))
- *Section 5.7.1, "Apache's Portable Runtime Library devel package renaming"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1247839))

A.5 2025-08-21

A.5.1 New

- *Section 5.5.2, "Support for Ansible interpreter"* (Jira (https://jira.suse.com/browse/PED-13352))

A.6 2025-01-09

A.6.1 New

- Added deprecation note for PHP 7.4 to *Section 9, "Removed and deprecated features and packages"* (Jira (https://jira.suse.com/browse/PED-8166))
- OpenLDAP server has been re-added to ease migration to 389 Directory Server. A note has been added to *Section 9, "Removed and deprecated features and packages"* to reflect that. (Jira (https://jira.suse.com/browse/PED-8118))
- *Section 5.12.2, "Effective user limits in systemd setup"* (Jira (https://jira.suse.com/browse/PED-7978))
- *Section 5.8.2, "`frr` (FRRouting Routing daemon) has been added"* (Jira (https://jira.suse.com/browse/SLE-13591))
- Added warning about changing system Python to *Section 5.5.3, "Alternative modern Python 3 interpreter upgraded to Python 3.11"*
- *Section 5.7.2, "`suseconnect -r` failure in Secure Boot mode"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1233354))

A.6.2 Updated

- Changed product names:
  - SLE-HA to SLE HA
  - Minimal-VM and Minimal-Image to Minimal VM and Minimal Image, respectively
  - SLE High Availability Extension to SUSE Linux Enterprise High Availability
  - SUSE Linux Enterprise-WE to SUSE Linux Enterprise WE
  - SLES-SAP to SLES for SAP
  - SLE-Software Development Kit to SLE Software Development Kit

A.7 2023-12-13

A.7.1 New

- *Section 5.7.4, "TigerVNC's `vncserver` has been removed"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1211550) ↗)

A.8 2023-11-06

A.8.1 New

- *Section 8.2.2, "Driver enablement for NVIDIA Grace"* (Jira (https://jira.suse.com/browse/PED-4564) ↗)
- *Section 8.4.1, "No graphics drivers on NVIDIA Grace Hopper"* (Jira (https://jira.suse.com/browse/PED-4564) ↗)

A.8.2 Updated

- Added NVIDIA Grace* to *Section 8.1, "System-on-Chip driver enablement"* (Jira (https://jira.suse.com/browse/PED-4564))
- Moved *Section 8.3, "64K page size kernel flavor is supported"* from *Section 2.8.1, "Technology previews for Arm 64-Bit (AArch64)"*, updating it and replacing it with *Section 2.8.1.1, "KVM virtualization with 64K page size kernel flavor"* (Jira (https://jira.suse.com/browse/PED-4489))

A.9 2023-09-29

A.9.1 New

- *Section 9.2.1, "Public Cloud module deprecations"* (Jira (https://jira.suse.com/browse/PED-3806))
- *Section 5.13.6.1, "Deprecating host network management with `libvirt`"* (Jira (https://jira.suse.com/browse/PED-5268))
- *Section 4.1.2, "SUSE Manager installation option not available"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1209235))
- *Section 4.5, "New Minimal VM and Minimal Image for IBM Z (s390x)"* (Jira (https://jira.suse.com/browse/PED-1911))
- *Section 2.8.2.1, "Support for AMD Wheat Nas GPU"* (Jira (https://jira.suse.com/browse/PED-983))
- *Section 8, "Arm 64-bit-specific changes (AArch64)"*
- *Section 2.8.1, "Technology previews for Arm 64-Bit (AArch64)"*
- *Section 8.2.1, "Driver enablement for NVIDIA Orin"* (Jira (https://jira.suse.com/browse/PED-1763))
- *Section 8.4.2, "No graphics drivers on NVIDIA Jetson"* (Jira (https://jira.suse.com/browse/PED-1763))

A.10 2023-05-22

A.10.1 Updated

- Changed wording of *Section 2.4, "Security, standards, and certification"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1211471)):

```
-SUSE Linux Enterprise Server 15 SP5 has been submitted to the certification bodies for:
-* Common Criteria Certification, see https://www.commoncriteriaportal.org/
-* FIPS 140-2 validation, see https://doi.org/10.6028/NIST.FIPS.140-2
+SUSE Linux Enterprise Server 15 SP5 will not be submitted for certification to either
 Common Criteria or NIST FIPS 140-3.
+As a practice, SUSE only submits even-numbered Service Packs (for example: SLES 15, SP2,
 SP4, etc.) for certification.
```

A.11 2023-05-15

A.11.1 New

- *Section 7.4.1, "Secure Execution"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1211316))

A.12 2023-05-11

A.12.1 New

- Added note about xpram removal in *Section 9, "Removed and deprecated features and packages"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1205381))
- *Section 6.3.3, "Adding changes in `ibmvnic` driver to assign IRQ affinity to all"* (Jira (https://jira.suse.com/browse/PED-2319))
- *Section 6.3.2, "`ibmveth` driver performance improvement"* (Jira (https://jira.suse.com/browse/PED-2638))
- Added note about NVMf-FC kdump in *Section 6.3, "Virtualization"* (Jira (https://jira.suse.com/browse/PED-1256))

- *Section 6.3.1, “Support for the new H_WATCHDOG hypercall included with P10 firmware”* (Jira (https://jira.suse.com/browse/PED-530))
- *Section 6.1.3, “POWER10 performance enhancements for cryptography: OpenSSL”* (Jira (https://jira.suse.com/browse/PED-512))
- Added a note about cryptography enhancements in NSS FreeBL in *Section 6.1, “Security”* (Jira (https://jira.suse.com/browse/PED-495))
- *Section 6.1.2, “POWER10 performance enhancements for cryptography: libgcrypt”* (Jira (https://jira.suse.com/browse/PED-520))
- *Section 6.1.1, “POWER10 performance enhancements for cryptography: nettle”* (Jira (https://jira.suse.com/browse/PED-505))

A.13 2023-05-10

A.13.1 New

- *Section 5.7.5, “`cpuid` has been added”* (Jira (https://jira.suse.com/browse/PED-2653))
- *Section 5.10.3, “dracut default persistent policy change”* (Jira (https://jira.suse.com/browse/PED-1884))
- *Section 7.4.2, “openCryptoki: `p11sak` support Dilithium and Kyber keys”* (Jira (https://jira.suse.com/browse/PED-2867))
- *Section 7.6.1, “KVM: Enable GISA support for Secure Execution guests”* (Jira (https://jira.suse.com/browse/PED-461))
- *Section 4.1.3, “Storage requirements for Btrfs installation on PowerPC”* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1209365))
- *Section 6.2.1, “NVMe-oF on LVM with multiple NVMe disks”* (Jira (https://jira.suse.com/browse/DOCTEAM-978))
- *Section 5.4.1, “`nouveau` disabled for Nvidia Turing and Ampere GPUs / openGPU recommendation”* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1210197))
- *Section 5.2.1, “Podman upgrade from 3.4.x to 4.3.1”* (Jira (https://jira.suse.com/browse/PED-1805))

A.14 2023-04-12

A.14.1 New

- *Section 7.7.5, "Add additional information to SCLP CPI"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.7.4, "Support Processor Activity Instrumentation Extension 1"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.7.3, "New CPU-MF Counters for IBM z16 and LinuxONE 4"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.7.2, "Support IBM z16 and LinuxONE 4 Processor-Activity-Instrumentation Facility"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.7.1, "NVMe stand-alone dump support"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.8.3, "Removed ESCON from installer"* (Jira (https://jira.suse.com/browse/PED-482) ↗)
- *Section 7.8.2, "Enabled installer to configure PCI-attached networking devices"* (Jira (https://jira.suse.com/browse/PED-454) ↗)
- *Section 7.8.1, "Auto scale `crashkernel` size with hardware changes"* (Jira (https://jira.suse.com/browse/PED-432) ↗)
- *Section 7.6.9, "KVM: Enablement of device busid for subchannels"* (Jira (https://jira.suse.com/browse/PED-3182) ↗)
- *Section 7.6.8, "Allow to list persisted `driverctl` definitions"* (Jira (https://jira.suse.com/browse/PED-2317) ↗)
- *Section 7.6.7, "KVM: Allow long kernel command lines for Secure Execution guests and QEMU"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.6.6, "KVM: Provide virtual CPU topology to guests"* (Jira (https://jira.suse.com/browse/PED-445) ↗)
- *Section 7.6.5, "KVM: Attestation support for Secure Execution (crypto), KVM: Secure Execution Attestation Userspace Tool"* (Jira (https://jira.suse.com/browse/PED-466) ↗)
- *Section 7.6.4, "Secure Execution guest dump encryption with customer keys, tool to process encrypted Secure Execution guest dumps"* (Jira (https://jira.suse.com/browse/PED-449) ↗)

- *Section 7.6.3, "`vfio-ap` enhancements"* (Jira (https://jira.suse.com/browse/PED-456) ↗)
- *Section 7.6.2, "Enhanced Interpretation for PCI Functions"* (Jira (https://jira.suse.com/browse/PED-473) ↗)
- *Section 7.5.6, "ROCE: Independent Usage of Secondary Physical Function"* (Jira (https://jira.suse.com/browse/PED-470) ↗)
- *Section 7.5.4, "`zipl`: Site-aware environment block"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.5.3, "`zipl` support for Secure Boot IPL and Dump from ECKD DASD"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.5.5, "Transparent PCI device recovery"* (Jira (https://jira.suse.com/browse/PED-457) ↗)
- *Section 7.5.2, "zdev: Site-aware device configuration"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.5.1, "Transparent DASD PPRC (Peer-to-Peer Remote Copy) handling"* (Jira (https://jira.suse.com/browse/PED-444) ↗)
- *Section 7.4.16, "`libica`: FIPS 140-3 compliance"* (Jira (https://jira.suse.com/browse/PED-2871) ↗)
- *Section 7.4.15, "`p11-kit`: add IBM specific mechanisms and attributes"* (Jira (https://jira.suse.com/browse/PED-440) ↗)
- *Section 7.4.14, "openCryptoki: PKCS #11 3.1 - support CKA_DERIVE_TEMPLATE"* (Jira (https://jira.suse.com/browse/PED-442) ↗)
- *Section 7.4.13, "`zcryptctl` support for control domains - kernel and s390-tools parts"* (Jira (https://jira.suse.com/browse/PED-480) ↗)
- *Section 7.4.6, "Display PAI (Processor Activity Instrumentation) CPACF counters (`s390-tools`)"* (Jira (https://jira.suse.com/browse/PED-472) ↗)
- *Section 7.4.12, "`libica`: eliminate SW fall backs - stage 2"* (Jira (https://jira.suse.com/browse/PED-468) ↗)
- *Section 7.4.11, "libica: extend statistics to reflect security measures (crypto)"* (Jira (https://jira.suse.com/browse/PED-478) ↗)
- *Section 7.4.10, "openCryptoki key generation with expected MKVP only on CCA and EP11 tokens"* (Jira (https://jira.suse.com/browse/PED-442) ↗)

- *Section 7.4.9, “openCryptoki: support crypto profiles”* (Jira (https://jira.suse.com/browse/PED-442))
- *Section 7.4.8, “openCryptoki EP11 token: vendor-specific key derivation”* (Jira (https://jira.suse.com/browse/PED-442))
- *Section 7.4.7, “openCryptoki EP11 token: IBM z16 and LinuxONE 4 support”* (Jira (https://jira.suse.com/browse/PED-442))
- *Section 7.4.5, “zcrypt DD: Exploitation Support of new IBM Z Crypto Hardware for kernel and `s390-tools`”* (Jira (https://jira.suse.com/browse/PED-472))
- *Section 7.4.4, “openCryptoki ep11 token: master key consistency”* (Jira (https://jira.suse.com/browse/PED-442))
- *Section 7.4.3, “In-kernel crypto: SIMD implementation of `chacha20`”* (Jira (https://jira.suse.com/browse/PED-439))
- *Section 7.3.1, “zlib CRC32 optimization for s390x”* (Jira (https://jira.suse.com/browse/PED-1350))
- *Section 5.5.3, “Alternative modern Python 3 interpreter upgraded to Python 3.11”* (Jira (https://jira.suse.com/browse/PED-3799))
- *Section 7.2.1, “Enablement for MIO Instructions - kernel and `rdma-core` parts”* (Jira (https://jira.suse.com/browse/PED-2175))

A.14.2 Updated

- Changed SLES release date year to the correct 2023

A.15 2023-03-01

A.15.1 New

- *Section 5.9.5, “Replacement of `gpg` as recommended tool for file encryption”* (Jira (https://jira.suse.com/browse/PED-1891))
- *Section 5.13.5.1, “open-vm-tools”* (Jira (https://jira.suse.com/browse/PED-1344))

A.15.2 Updated

- *Section 5.13.2, "Xen"* (Jira (https://jira.suse.com/browse/PED-1858))
- *Section 5.13.3, "QEMU"* (Jira (https://jira.suse.com/browse/PED-1716))
- *Section 5.13.1, "KVM"* (Jira (https://jira.suse.com/browse/PED-1855))
- *Section 5.13.4, "libvirt"* (Jira (https://jira.suse.com/browse/PED-447), Jira (https://jira.suse.com/browse/PED-225))

A.16 2023-02-01

A.16.1 New

- *Section 4.1.4, "Installation via SSH on s390x"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1206585))
- *Section 5.12.3, "Silence `KillMode=None` messages"* (Jira (https://jira.suse.com/browse/PED-407))

A.17 2022-11-30

A.17.1 New

- Deprecation of `gnote` in *Section 9.2, "Deprecated features and packages"* (Jira (https://jira.suse.com/browse/PED-1839))
- Removal of `samba-ad-dc-libs` in *Section 9.1, "Removed features and packages"* (Jira (https://jira.suse.com/browse/PED-143))
- *Section 5.12.4, "`yast2-iscsi-client` drops `open-iscsi` and `iscsiuio` as dependencies"* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1204978))

A.18 2022-11-02

A.18.1 New

- *Section 5.6.2, “Restoring default Btrfs file compression”* (Jira (https://jira.suse.com/browse/PED-63) ↗)
- Added note about `openmpi2` and `openmpi3` in *Section 9.2, “Deprecated features and packages”* (Jira (https://jira.suse.com/browse/PED-904) ↗)

A.19 2022-10-18

A.19.1 New

- Added note about removing Kerberos/LDAP from YaST in *Section 9, “Removed and deprecated features and packages”* (Bugzilla (https://bugzilla.suse.com/show_bug.cgi?id=1202257) ↗)
- Added note about removing `thunderbolt-user-space` in *Section 9, “Removed and deprecated features and packages”* (Jira (https://jira.suse.com/browse/PED-1358) ↗)

A.19.2 Updated

- Updated Java lifecycle in *Section 5.5.5, “Supported Java versions”* (Jira (https://jira.suse.com/browse/PED-1590) ↗):
  - OpenJDK 11 end of life is now end of 2026
  - OpenJDK 17 added
  - OpenJDK 18 end of life is now end of 2026

B Kernel parameter changes

> **Warning**
>
> This list of changes may not be exhaustive.

B.1 Changes from SP4 to SP5

These Linux kernel parameters have been changed since SLES 15 SP4.

Setting	Previous value	New value	Caused by package	More details
`sysctl_net_core_bpf_jit_limit`	264241152	528482304	kernel	Bug 1218234 (https://bugzilla.suse.com/show_bug.cgi?id=1218234)
`sysctl_net_ipv6_route_max_size`	4096	2147483647	kernel	Bug 1219295 (https://bugzilla.suse.com/show_bug.cgi?id=1219295)